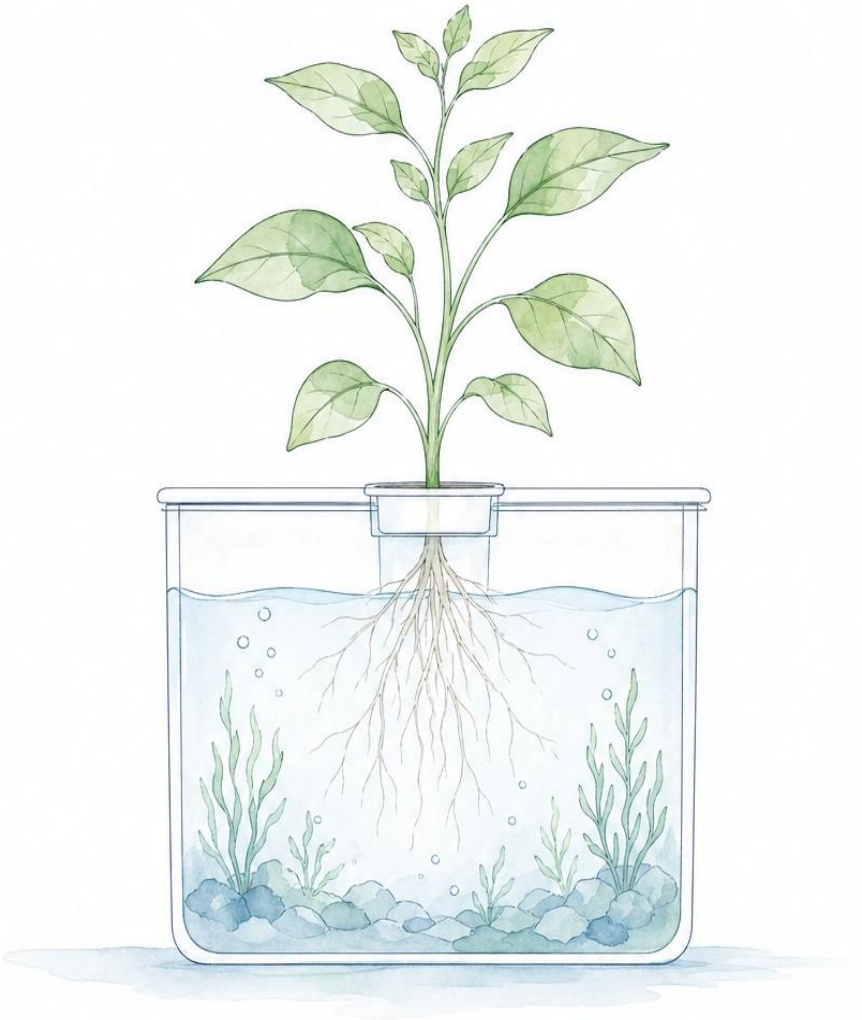
BIONEXIS

Team No. 30207

2026/2025



BIONEXIS

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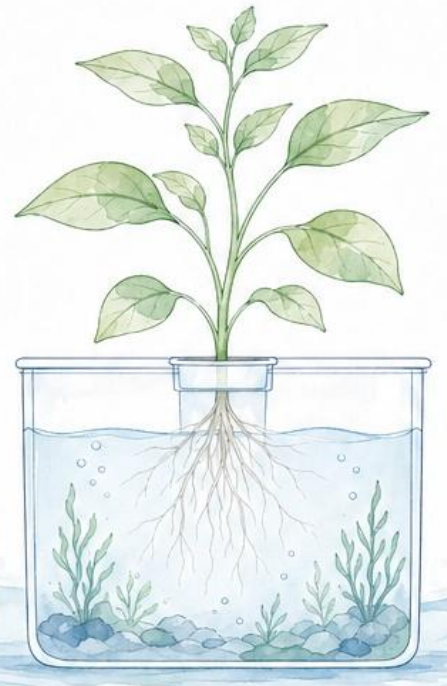


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Introduction

With the constant growth of our motherland, it faces progressively harder obstacles and more challenging problems. These problems are usually referred to as “The 11 Egyptian Grand Challenges.

Nowadays, Egypt stands at a critical crossroads, facing a set of interconnected problems and challenges that will shape its future. The continued population growth, without proper land and resource usage, alongside the constant need for urbanization and staggering economic ambitions, poses a significant threat.

All of this takes place while Egypt struggles to cope with new emerging problems and challenges, such as global warming, which has recently affected the world, or the need to adopt new and improved processes and methods for producing energy instead of the traditional carbon-emitting means and methods used to generate electricity and energy. These Grand challenges represent the key obstacles that Egypt faces in the long-term run and the sustainability of the country.

Each challenge requires innovative solutions, strategic planning, and collective efforts from everyone in the country, including the government, which should support the project's efforts, as well as politicians, business owners, scientists, and citizens alike, even students like us. Each person's effort is important for our beloved country, paving the way for a stronger, greener, and more resilient future.

The first step to solving the problem is to know what the problem is, which is why we have mentioned the 11 Grand Challenges that Egypt faces, and to solve them, making Egypt more developed:

- Improve the use of alternative energies.
- Recycle garbage and waste for economic and environmental purposes.
- Deal with urban congestion and its consequences.
- Work to eradicate public health issues/diseases.
- Increase the industrial and agricultural bases of Egypt.
- Address and reduce pollution fouling our air, water, and soil.
- Improve the use of arid areas.
- Manage and increase the sources of clean water.
- Deal with population growth and its consequences.
- Improve the scientific and technological environment for all.
- Reduce and adapt to the effects of climate change

Chapter I

(Present and Justify a Problem and Solution Requirements)

I. Egypt Grand Challenge

II. Problem to be solved

III. Research

IV. Prior solution

Egypt Grand Challenge:

❖ Address and reduce pollution fouling our air, water, and soil:

The term "pollution" means: the presence of a substance in or into the environment which has harmful or hazardous effects, according to the Oxford Languages dictionary.
Pollution could be seen as a key player in shaping the sad reality of our beloved country, as more pollution means fewer available resources, which could make Egypt vulnerable to some serious problems, such as:

- **Regional disasters:** Famines could initiate as less farmable lands mean fewer crops, leading to food shortages in developing countries such as Egypt.

- **Health hazards:** As…

- Air pollution is a primary cause of immunity deficiency and increases the chance of catching bodily issues like pulmonary cancers and other pulmonary diseases like influenza, and allergies due to the inhalation of polluted air containing CO_2, CO, and other harmful gases.

- Anybody could consume contaminated water that contains a harmful substance like lead or arsenic, and maybe even a harmful micro-organism like malaria plasmodium or bilharzia.

- **Economic problems**: it also harms the economy as it could affect our valuable resources like water, air, and the fertile soil used for producing essential crops, leading to a subtle increase in the prices of our basic products, and this could lead to potential wars about something we take for granted, like wheat.

- **Global warming & Climate change:** As **we** all know, greenhouse gases like CO_2, CH_4, and other gases presented in Figure 3, which are caused primarily by air pollution, form a

layer around the earth that allows white light to pass through it, but traps infrared rays that cause the gradual rise in the average temperature of the earth.

This increase causes various problems, like:

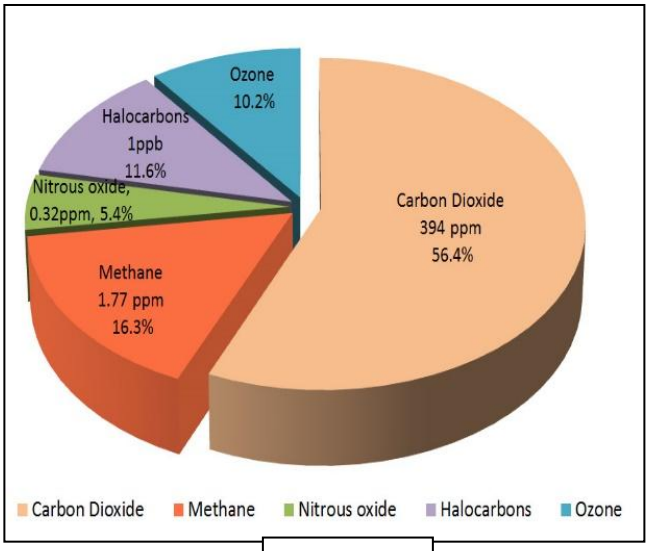


Chart 1

- **Melting ice caps**:
  The polar ice caps melt as the temperature increases, causing the rise of the ocean level, leading to some low-lying regions like Alexandrea to sink underwater.

- **Climate change:**
  As the average temperature of the globe is constantly rising, the weather becomes more extreme as winters become colder and summers become hotter. This affects the environment and the ecosystem.

Some arctic animals, like penguins, cannot survive in the hot biomes or in hot weather, and African Elefants cannot endure the harsh winters. This shift could result in the extinction of some animals or even whole species.
Some crops need special requirements to grow, like rice, needing excessive amounts of water and sunlight to grow. If the climate shifts slightly, this could make the biome that used to grow these crops incapable of producing them, leading to the problems mentioned before.

In summary, pollution causes various problems like health hazards and famines, poses a potential economic threat, and endangers the global ecosystem. A problem with a horrendous impact like this needs to be stopped, no matter the cost.

❖ Recycle garbage and waste for economic and environmental purposes:

The environmental necessity of waste management and recycling regarding its economic effect. Waste generation is an unavoidable product of human civilization, connected deeply to the social and economic aspects of civilians' lives.
As global economies transition from linear to circular models, the effective management, recovery, and recycling of waste streams have become critical for both environmental and economic sustainability. Effective action is necessary to achieve ambitious goals, such as the

EU's 55% target for plastic packaging waste recycling by 2030, which addresses high dependence on fossil feedstock, low recycling rates, and significant leakage of plastics into the environment. Developing innovative technologies like chemical and combined physical recycling is key to securing material resources, increasing recycling rates, and potentially improving environmental performance, especially by handling waste flows otherwise sent to disposal or incineration.

The Environmental Performance and Life Cycle Assessment of Waste Management:

Environmental assessment tools, specifically Life Cycle Assessment (LCA), are employed to evaluate the performance of Municipal Solid Waste (MSW) management systems and to quantify the potential environmental impacts resulting from different waste treatment methods. LCA models investigate waste management options by computing impact levels based on detailed unit processes, quantified values of various resources, and emissions.

For waste management systems, the choice of preferred management option is based on three main criteria. The first of these is the maximization of material recovery while minimizing processing impacts, primarily related to energy consumption. This criterion directly reflects environmental performance and is in line with the waste hierarchy.

Comparison of Recycling and Energy Recovery Studies comparing recycling technologies (mechanical, physical, chemical) with energy recovery (incineration) for specific waste streams, such as plastic waste, reveals several critical findings:

- **Climate Change (Global Warming Potential - GWP):**
  Recycling is generally preferable to energy recovery in all pathways analysed for climate change, notably for mixed polyolefin waste that is not currently mechanically recycled. This is because the environmental savings from energy recovery are often insufficient to compensate for the environmental impacts of waste incineration and its associated emissions. For example, the management of one tonne of mixed shredded plastics from large WEEE (cooling and freezing appliances) resulted in net savings for mechanical recycling (-1522 kg-eq./t) but net burdens for energy recovery via incineration (1095 kg -eq./t).

- **Other Environmental Categories (Acidification and Particulate Matter):**
  While maximizing material recovery holds for categories like ozone depletion and fossil resource use, for some categories, including acidification, particulate matter, and eutrophication, energy recovery can perform better than energy-intensive recycling pathways. This counter-performance is largely because current incinerators in Europe, on average,

achieve much lower emissions of substances like dust than the average energy production mix used for calculating the impacts of recycling's energy consumption. The shift to a cleaner European energy mix is expected to further increase the gap in favour of recycling, supporting the validity of the EU waste hierarchy.

- **Municipal Solid Waste (MSW) Management:**

  In the context of MSW, which includes collection, transportation, and disposal, a "Conjunctive Disposal" scenario (composting and landfilling) was found to yield minimum environmental impacts compared to a "Business as Usual" scenario (collection, transportation, landfilling) and a "Recycling" scenario, based on impact metrics including GWP, Acidification Potential (AP), and Eutrophication Potential (EP) Economic Viability and Technological Imperatives.

  The decision on the preferred waste management option must also consider the technical feasibility (specificity of the plastic waste stream and treatment required) and the economic feasibility.

- **Economic Factors in Plastic Recycling**

  The economic viability of physical and chemical recycling, particularly without public support, is determined by three main parameters:

  feedstock prices, capital expenditures (CAPEX) and operational expenditures (OPEX), and output prices.

- **Cost and Revenue:**

  Preliminary economic data suggests some physical and chemical recycling technologies, such as dissolution and glycolysis, may have already reached break-even status in terms of costs, although other technologies like metanalysis, pyrolysis, and gasification were found to have negative net incomes. However, this current analysis does not account for possible gate fees or other cost correction measures.

- **Future Projections:**

  Physical and chemical recycling technologies are still experiencing rapid development, and their costs are projected to decrease in the future. Simultaneously, costs for mechanical recycling are expected to remain stable, while costs for virgin plastics production are expected to substantially increase with increasing fossil fuel prices. This shifting in the economic landscape is estimated to lead to positive earnings for major chemical recycling technologies, like metanalysis by 2025, pyrolysis by 2033, and gasification by 2040.

- **Cost Efficiency and Alternative Energy Recovery:**
  The environmental ranking of waste management scenarios can often correlate with economic performance. For MSW management in Kathmandu, the
  A conjunctive disposal scenario (composting and landfilling) that yielded minimum environmental impacts was also identified as a cost-efficient option.
  Furthermore, waste management offers a pathway to sustainable energy recovery. The anaerobic digestion of organic waste combined with livestock manure, which is part of effective waste management, is highlighted as a
  sustainable and environmentally friendly energy option for producing biogas. Fuel Energy Consumption (FEC) itself is a key metric, calculated based on the monetary value expended to purchase fossil fuels (petrol, diesel, and kerosene) used in the collection, transport, and management of waste.

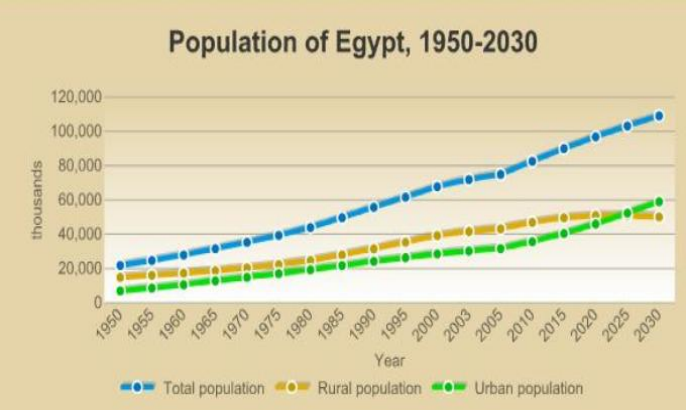
  In conclusion, the effect of waste recycling serves the double purpose of maximizing material recovery—aligning with the waste hierarchy and achieving environmental savings in climate change—while simultaneously creating economically viable industrial sectors projected to reach profitability in the medium to long term, thereby fostering a robust circular economy.

❖ Deal with population growth and its consequences.

The largest issue with Egypt and to this day is population growth. It halts the positive outcomes of sustainable development schemes and the state's efforts to ensure that people live better lives, particularly when the new republic is struggling to make Egyptians' lives easier. The increasing population enormous burden on the state budget. Although the state has made great efforts since the 1930s to overcome overpopulation, it still cannot address the vast growth. The developed plans are not extensive and stringent enough, and they do not slow down the rate of growth to align with the amount of state resources and sustainable development objectives.

Population growth is not a big issue, as evidenced by the fact that the number of children per woman is declining. In Egypt, the birth rate of women was 7.2 in the early sixties and 3.4 in 1998. The United Nations predicts that this figure will reduce to three in 2000-2005 and to two in 2020-2025.

However, a closer look reveals that the increase of the population remains a problem in the minds of the Egyptian policymakers. Should the reduction in births continue to trend and the aim of the government is to achieve a replacement level of births by 2016, the government would require an effective family-planning program that provides quality services to more individuals. This program will be funded more and more by the resources of the Egyptian government itself.



Graph 1

The graph will be used to demonstrate the number of individuals who will live in Egypt between 1950 and 2030. It divides the figures into three stages, namely total population, people living in rural areas, and people living in urban areas. Through these trends, we are able to know the way the population of Egypt has been evolving and what such an increase may imply for the future of Egypt as a nation.

Total Population Growth:

The overall population has been on a constant rise. Between 1950 and 1970, it increased by a small margin per year, primarily due to the fact that a lot of babies were born and fewer have died, owing to improved health care and vaccinations. From 1970 to 2000, the growth got faster. The number increased to nearly 70 million, as compared to some 36. The health care, clean water, and food were improved, and made people live longer. The growth accelerated further after 2000. Egypt had over 100 million people by 2020, with the numbers expected to increase to 110-115 million by 2030. This rapid increase puts a strain on the government to develop more houses, roads, health care, schools, and jobs for the increasing population.

Rural Population Trends:

The countryside population has continued to increase since it had 13 million in 1950 and nearly doubled by 47 million in 2030. When the majority of Egyptians were living in the villages, they typically engaged in agricultural labor. They also had families that were larger, and thus, the birth rate was high in villages. Although the figure in the rural areas continues to increase, the rate is not as high as in urban areas. The main cause of the latter is the fact that the majority of the population is moving out of the countryside to the city in search of high-quality working places, education, health services, and general standards of living. Despite the increased migration, there are real issues for people in the rural regions, such as a lack of good medical services, poor schools, a lack of jobs, and poverty. With an increasing population, farms, water, and food production might become strained, increasing the likelihood that there may be those whose food needs will not be met.

Urbanization and Urban Population Growth:

Among the trends that are most evident in the graph, there is the rapid increase in the number of urban dwellers. Only an approximate of 7 million Egyptians were residing in cities in the year 1950, whereas this will be very near 60 million in the year 2030. This massive migration demonstrates that a large number of people are shifting to towns and cities. These places, including Cairo, Giza, Alexandria, and other key cities, have expanded significantly over time. Individuals travel to cities with the hope of employment, increased salaries, improved education, and increased medical attention. This city has the potential to boost the economy, but it generates severely challenging issues. The increasing urbanization of cities has led to congestion, road congestion, a lack of housing, and the proliferation of informal settlements (slums). There is an inadequacy of water and sewage, roads, and garbage services in cities in various locations. Such issues reduce the quality of life of people and may cause problems with health and the environment.

Economic Consequences of Population Growth:

One of the most important consequences of rapid population growth is its economic impact. On one hand, a growing population provides a large labor force, which can support economic production and development. On the other hand, if economic growth does not keep pace with population growth, unemployment and poverty can increase. In Egypt, population growth has created significant pressure on the job market. Each year, millions of young people enter the workforce, but job creation has not always been sufficient. This has led to higher unemployment rates, especially among youth, and has increased dependence on informal employment. Moreover, the government must allocate large portions of the national budget to essential services such as education, healthcare, housing, and subsidies, which limits investment in infrastructure and industrial development. As a result, economic progress becomes more challenging.

Environmental consequences:

The increase in population is destructive to the environment. As the population increases, they require increased water, food, energy, and land. Egypt is already struggling to acquire water since they channel on the Nile River. An increase in population equates to increased stress on the little water that is available. This may complicate farming processes and make daily needs impossible to fulfil.

Since urbanization means settlement in urban areas, it deprives the country of agricultural land, particularly in the fertile Nile Delta. Due to this fact, we produce less food in our local places and must purchase food elsewhere. We also generate more trash, trucks and cars are polluting the air, and the factories emit pollutants. All these worsen the environment, leading to climate issues. Unless we employ clever means to expand and retain the environment, we can readily damage it, making it detrimental to the generations to come.

Increase the industrial and agricultural bases of Egypt:

In the recent years, Egypt has confronted genuine difficulties due to normal asset botch, natural corruption, contaminating modern generation techniques and exponential increment in unsustainable samples of utilization. Egypt is confronting remarkable asset emergency, particularly in vitality, water and to a lesser degree sustenance. are cooperating with the Egyptian Ministry of environment and therefore the Center for Environment and Development for the Arab Region and Europe (CEDARE) to form a move build up a maintainable utilization and generation (SCP) national activity design. The approach creators chose that ought to be attended critically incorporates farming, vitality, water and civil strong waste. during this story we'll absorb more on how Egypt is moving, gradually however definitely, towards manageable agrarian practices.

Reasons for agriculture:

1. Egypt has minimal successful precipitation, at the most 200 mm and unequally circulated and on restricted territories.

2. Over the previous decades, repetitive dry spell, current innovation and new monetary principles have profoundly changed sheep creation frameworks and financial conditions. Expanding the itinerant settlement, expanding the number of sheep in minor territories and expanding exploitation and lessening have enormously expanded the load on accessible land and decreased soil fruitfulness.

3. Absence of enthusiasm for farming work and absence of excellent living conditions and great salary.

4. Also, lack of agricultural production and lack of animal food results in inaccessibility of mechanical crude materials, shortage of animal products and causing weakness of the Egyptian economy significantly.

Impacts of this problem:

1. Pesticides that are utilized to disposal of unsafe creepy crawlies, microorganisms and different irritations which blend with soil, water, air and nourishment, cause issues on the

horticultural sustenance's and influence both human wellbeing and customary adjust so finally they become an environmental problem.

2. The compost that contains abnormal amounts of nitrogen which are utilized to reinforce plant development, increasingly and qualified items and a couple of highlights of soil like physical, synthetic and body cause natural contamination if there should be an event of unreasonable or wrong utilization like groundwater contamination due to appearance of huge amounts of nitrogen in water. This additionally influences the water living beings and when that kind of waters wont to some place they break the regular adjust of condition.

3. Water systems have enormous significance to high agrarian yield and quality arid and semi-arid regions. Be that because it may, wrong water system causes condition issues. Ascending of spring water saltiness, manures and substance added substances buildups dive too deep with water system water, follow components take in water sources and cause soil disintegration and these kinds of waters make sickness and unsafe beat all living creatures, so this type of water is an important situation issue. Likewise, unreasonable water system as a reason for farming creation prompts soil saltiness and desertification.

Industrial Sector:

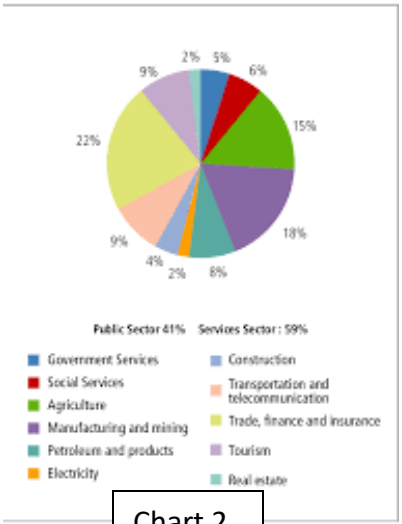


Chart 2

The industrial sector may be a major growth driver having strong backward and forward linkages with both the agriculture and services sector. it's required to assume a viable part in reactivate financial development within the Egyptian economy over the medium and long terms. Industry represented 30% of Gross domestic product in 2001. Major modern items included materials, chemicals (counting manures, polymers, and petrochemicals), pharmaceuticals, nourishment preparation, oil, development, concrete, metals, and lightweight buyer products. The apparel and materials division are the most important mechanical managers. More noteworthy Cairo, Alexandria, and Helwan are Egypt's primary mechanical focuses, delivering iron and steel, materials, refined oil-based goods, plastics, building materials, gadgets, paper, trucks and cars, and chemicals. The Helwan iron and 17 | Page steel mill, 29 km (18 mi) south of Cairo, utilizing imported coke, forms press metal mined close Aswan into sheets, bars, billets, plates, and blossoms. Farming, vehicle, buyer hardware, Development part, material, nourishment preparing, tourism, hydrocarbon, and development, light fabricates guard industry.

Industry assumes an important part in: -

- Providing employment opportunities.
- Diminishing joblessness issues.
- The development of varied economic activities
- Diminishing reliance on imports and supporting the fare limits of the state.
- Raising the expectation for everyday comforts of individuals.

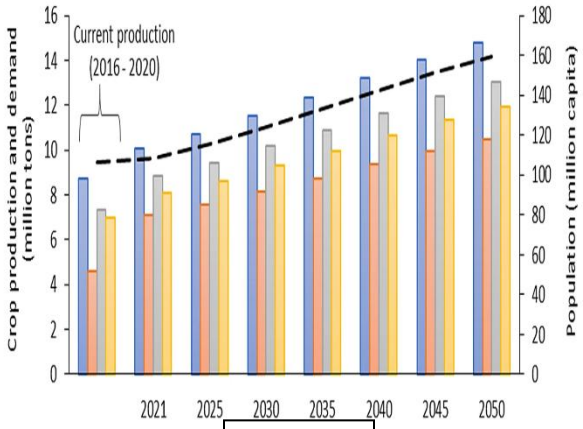
II. Problem to Be Solved:

The population growth is the most serious challenge facing Egypt today. Over past years, Egypt population increased rapidly that's make pressure on the Egypt resources and services and this lead problems in the environmental, economic, and social and this lead to that many people does not have jobs and this lead to have problems in money and this lead to have problems in food and not only this and the uncontrolled growth lead to overcrowded school if this problem solved life will be better.

If the problem was solved

❖ An expected surge in the industrial and the agriculture base:

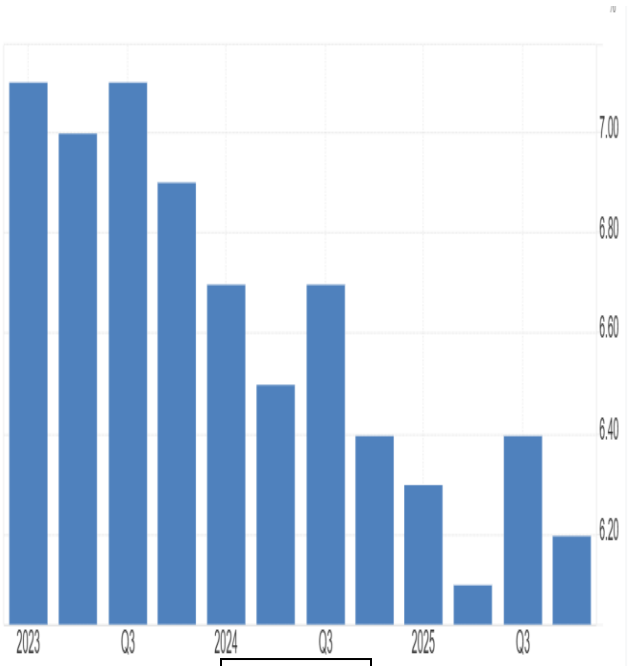
When the population growth is controlled, Egypt will spend money on projects in industries like factories. Agricultural like less pressure on water, and this leads to an increase in food, which would reduce the need to import food and help to achieve food security, and vegetables instead of spending it on the economic and healthcare problems that are caused by population growth.



Graph 3

❖ Unemployment rates will decrease:

If the population growth is controlled properly in Egypt unemployment rates will decrease gradually for serval reasons. When the population increases the economy cannot always create jobs for all of them but if it is controlled the number of job seekers will be balanced with number of available jobs and not only this when the population growth is controlled the government will be able to invest economic developments projects instead of spending it in the over used services and will invest in industrial like factories and small businesses which create more jobs and this good planning improve education system and adults when graduate with better skills that will match requirements of the job that will gradually decrease unemployment.



Graph 2

III. Research:

Thorough research was conducted on the topics related to the project in order to generate the most optimal solution possible. These topics were:

- Various plants species are based on their optimal environmental conditions.
- Electric components, Sensors, actuators, and microcontrollers.
- Various aquatic creatures regarding their optimal environmental conditions.
- Aquaponic System Designs.
- Techniques for plastic recycling and how used plastic is managed.

1) Plant Species (optimal conditions):

Lettuce (Lactuca sativa)

1. Overview



Figure 1

- Family is Asteraceae
- The plant type is a vegetable
- Lettuce comes from the Mediterranean region
- It takes 30 to 45 days for Lettuce to grow

Lettuce is a common crop in vertical farms all over the world because Lettuce grows quickly and has a compact structure.

---

2. Biological Characteristics

- Lettuce has a shallow root system
- Lettuce has a water content which is about 95 percent
- Lettuce grows well in cool conditions
- Lettuce is harvested as whole heads or loose leaves

Some common varieties of Lettuce are

- Romaine Lettuce
- Butterhead Lettuce
- Iceberg Lettuce

---

3. Why Lettuce is Ideal for Smart Farms

The key advantages of Lettuce are

1. Lettuce has a fast growth cycle which's about 3 to 5 weeks
2. Lettuce has a root system which is ideal for hydroponics
3. Lettuce requires light compared to fruit crops
4. Lettuce gives a yield per square meter

Vertical farms can produce up to 20 times more Lettuce per area than traditional farming because of stacked layers and controlled conditions.

---

4. Growing Requirements

The range for Lettuce is

Factor Optimal Range

- Temperature is 15 to 20 degrees Celsius
- pH is 5.5 to 6.5
- Light is 12 to 16 hours of LED
- Water system is Hydroponic NFT or aeroponics

---

5. Smart Farm Technologies Used

Some technologies used for Lettuce are

- LED spectrum control which includes red light
- Nutrient film technique which is NFT
- IoT sensors for humidity and nutrients
- Automated harvesting robots

---

2. Basil (Ocimum basilicum)

1. Overview

- Family is Lemnaceae
- The plant type is a herb
- Basil comes from Tropical Asia
- It takes 25 to 40 days for Basil to grow

Basil is one of the profitable herbs in vertical farming because Basil has a high market price and can be continually harvested.



Figure 2

---

2. Biological Characteristics

- Basil has leaves which are rich in essential oils
- Basil is a fast-growing herb
- Basil can be harvested continuously

Some popular types of Basil are

- Sweet Basil
- Thai Basil

---

3. Advantages of Smart Farming

The advantages of Basil are

- Basil has an economic value
- Basil is a small plant
- Basil can be harvested continuously which is a cut-and-come-again crop

Herbs like Basil can be harvested repeatedly from the same plant for several weeks.

---

4. Growing Requirements

The range for Basil is Factor Optimal Range

- Temperature is 22 to 28 degrees Celsius
- pH is 5.5 to 6.5
- Light is 14 to 16 hours
- Nutrients are high in nitrogen

---

5. Smart Farm Applications

Some technologies used for Basil are

- AI monitoring for leaf health
- Precision nutrient dosing
- Automated pruning systems

---

3. Strawberries (Fragaria × ananassa)

1. Overview

- Family is RosaceaeThe plant type is a fruit crop
- It takes 60 to 90 days for Strawberries to grow

Strawberries are seen as a high-value fruit for farming and are increasingly grown in smart farms.



Figure 3

---

2. Biological Characteristics

- Strawberries are plants grown as an annual crop
- Strawberries produce flowers and fruits at the same time

- Strawberries are sensitive to climate variations
- Day-neutral varieties are commonly used for Strawberries.

---

3. Advantages of Smart Farms

farms provide

- Controlled pollination for Strawberries
- Stable fruit production for Strawberries
- Year-round harvesting for Strawberries

Hydroponic systems can yield about 12.5 kilograms per square meter in vertical setups for Strawberries.

---

4. Growing Requirements

The optimal range for Strawberries is Factor Optimal Range

- Temperature is 18 to 24 degrees Celsius
- pH is 5.8 to 6.2
- Light is 14 to 18 hours
- Pollination's done by bees or robotic pollinators

---

5. Smart Technology

Some technologies used for Strawberries are

- AI disease detection
- harvesting
- Computer-controlled fertigation

---

4. Cherry Tomato (Solanum Lycopersicon)



Figure 4

1. Overview

- Family is Solanaceae
- The plant type is a fruit vegetable
- It takes 60 to 85 days for Cherry Tomato to grow

Compact tomato varieties like Cherry Tomato and grape tomatoes are commonly grown in systems.

---

2. Biological Characteristics

- Cherry Tomato is a vine plant
- Cherry Tomato requires support or trellis
- Cherry Tomato produces clusters of fruits

---

3. Advantages for Smart Farms

The advantages of Cherry Tomato are

- Cherry Tomato has a consumer demand
- Cherry Tomato gives a good yield in controlled environments
- Cherry Tomato produces fruits continuously

---

4. Growing Requirements

The range for Cherry Tomato is Factor Optimal Range

- Temperature is 20 to 26 degrees Celsius
- pH is 5.5 to 6.5
- Light is high intensity LED
- CO_2 is often enriched to increase yield for Cherry Tomato

---

5. Smart Farm Technologies

Some technologies used for Cherry Tomato are

- CO_2 enrichment systems
- Automated trellis support
- Climate control algorithms

---

5. Microgreens (Radish / Broccoli / Pea Shoots)

1. Overview

Microgreens are vegetable seedlings harvested after 7 to 14 days.

Some examples of Microgreens are

- microgreens
- Broccoli microgreens
- Pea shoots



Figure 5

These crops are widely used in restaurants and health food markets.

---

2. Biological Characteristics

- Microgreens are small plants
- Microgreens are harvested at the seedling stage
- Microgreens are extremely nutrient-dense

---

3. Advantages for Smart Farms

The advantages of Microgreens are

- Microgreens have a short growth cycle which is about 1 to 2 weeks
- Microgreens have a high nutrient concentration
- Microgreens require minimal space

Microgreens often produce high profits per square meter in vertical farming.

---

4. Growing Requirements

The range, for Microgreens is Factor Optimal Range

- Temperature is 20 to 24 degrees Celsius
- pH is 5.5 to 6.5
- Light is moderate LED
- Harvest is done in 7 to 14 days for Microgreens

---

5. Smart Technology

Some technologies used for Microgreens are

- Automated seed spreading
- Computer-controlled irrigation
- AI harvest timing

2. Electric Components, Sensors, Actuators, and Microcontrollers:

- ## Electric Components

Electrical devices have common components that participate in performing specific functions of this device; all these components are involved in the electric circuit of this device. These components are:

1. **Resistor:**

   A resistor is one of the most fundamental electrical components. It resists the flow of electric current, limiting or controlling the amount of current that passes through a circuit.



Figure 6

   - **Function:**

     Resistors are primarily used to control voltage and current levels within a circuit. They dissipate energy in the form of heat and help protect sensitive components from receiving too much current, which could cause damage.

---

2. **Capacitor:**

   A capacitor is a device that stores electrical energy in an electric field. It temporarily holds electric charge, then releases it when needed.
   Measuring unit: Farad.

   - **Function:**

     Capacitors are used for storing energy, filtering signals, and stabilizing voltage and power flow. They smooth out fluctuations in electrical signals and provide filtering in power supplies to remove unwanted noise.



Figure 7

---

3. **Inductor:**

   Inductors are passive components that store energy in the form of a magnetic field when current flows through them.
   Measuring unit: Henry

   Function:

   Inductors oppose changes in current, making them useful for filtering high-frequency signals or storing energy in devices like transformers or motors. They can smooth out the ripple in power supply circuits and are



Figure 8

often used in combination with capacitors to form tuned circuits.

---

4. Diode:

A diode is a semiconductor device that allows current to flow in one direction only, blocking current in the opposite direction



Figure 9

- **Function:**
  Diodes are primarily used for rectification, converting alternating current (AC) to direct current (DC). They are also used to protect circuits by preventing reverse current, which can damage sensitive components.

---

5. Relay:

A relay is an electrically operated switch that uses a small current to control a larger current in another circuit. It allows one circuit to control another without a direct electrical connection between the two.



Figure 10

- **Function:**
  Relays are commonly used for electrical isolation between high and low-voltage circuits. They also protect by allowing circuits to handle high-power loads that a simple switch might not handle safely.

---

6. Bridge Rectifier:

A bridge rectifier is an arrangement of four diodes connected in a bridge configuration. It converts alternating current (AC) into direct current (DC).

- **Function:**
  A bridge rectifier allows current to flow in only one direction, effectively transforming AC into DC. It is a key component in power supply circuits used in chargers and adapters.



Figure 11

❖ Electronic devices:

An electronic device is a gadget employing semiconductive elements, such as silicon, Germanium, and the like, to facilitate electrical flow and execute specific functions. Take a mobile phone, for instance; its circuitry is predominantly composed of semiconductor materials. Upon activation, the mobile phone adeptly transforms electrical energy into intelligent operations, encompassing touch interactions, gesture control, networking capabilities, and memory management**.**

Electronic devices, on the other hand, exclusively function with Direct Current (DC). The voltage variety for digital gadgets is typically within the 'mV' to 'V' range, with common ratings like 5V, 12V, 24V, and many others.

Electronic gadgets, in assessment, consciousness on controlling voltage and present day. This management is accomplished through using lively components like transistors and diodes, taking into consideration the manipulation and amplification of electrical signals.

Electronic gadgets, exemplified by using modern technology like cellular phones, have the capacity to make short choices. This ability for decision-making units, digital devices aside, in terms of functionality. So, it has common parts with electric devices, with some little differences between them, which are:

---

1. Integrated Circuit (IC):

An IC is an electronic component that contains many connected circuits inside a single package, sometimes called a chip. It often has a 3- to 5-digit number printed on it (plus letters) that identifies its function or manufacturer.

- **Function:**



Figure 12

ICs perform a wide variety of tasks depending on their type: power regulation (e.g., 7805, LM317), logic operations (e.g., CD4026, CD4017), signal generation (e.g., NE555, LM741), switching (e.g., CD4066), or driving circuits (e.g., ULN2003). They provide compact, efficient, and reliable circuit implementations by integrating many components.

•Package Types:
– DIP (Dual In-Line Package), e.g. DIP8 has 8 pins.
– Metal-can (e.g., TO-99), often used for high-quality op-amps or precision ICs

- **Microcontrollers**

- **Arduino Uno:**

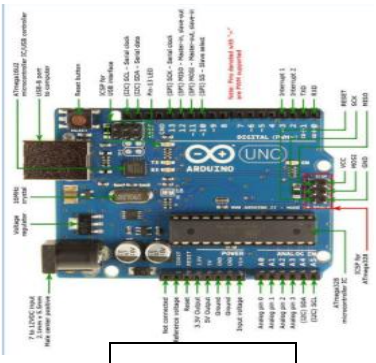


Figure 13

The microcontroller exhibited in Fig. 1 is an open-source microcontroller that can be easily programmed and updated at any time. Built with the 8-bit Atmel AVR microcontrollers that are manufactured by Atmel or a 32-bit Atmel ARM, these microcontrollers can be programmed easily using the C or C++ language in the Arduino IDE. When paired with a Wi-Fi module (e.g. ESP8266), it can send data to a local website.

The Arduino consists of many components, as shown in Figs 1&2, such as:

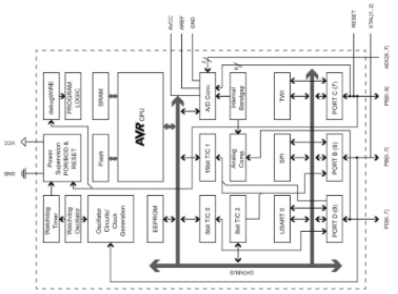


Figure 14

- Microcontroller (ATmega328): This is the brain of the development board that can receive, store, and send information or commands to the peripheral devices
- Analog I/O Pins: Pins A0-A5 can read the signal from an analog sensor.
- Digital I/O Pins: Pins 0-13 can be used for digital input or output.
- TX/RX: Pins transmit and receive data.

- **ESP32:**

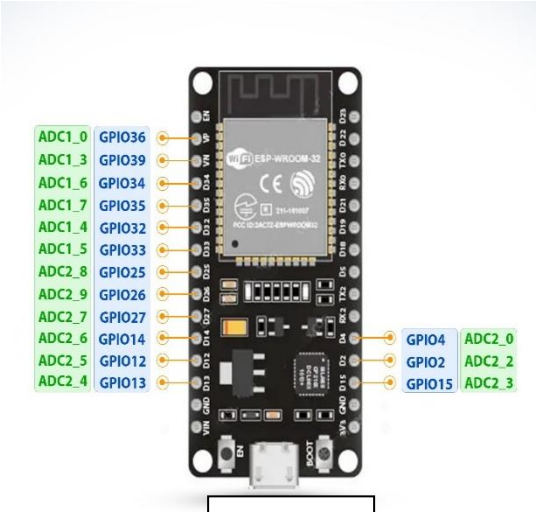


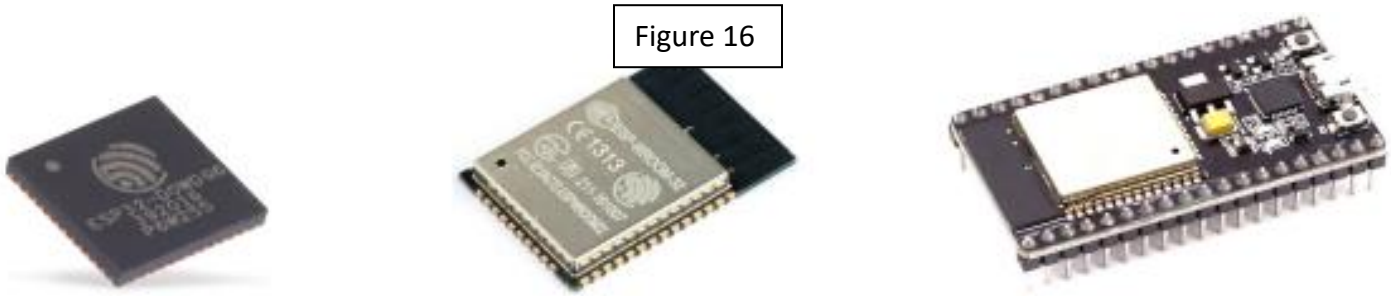
Figure 15

ESP32 (2016) is a hybrid Wi-Fi and Bluetooth SoC that integrates Tensilica Xtensa LX series 32-bit RISC processor with a TCP/IP stack. As a successor of the ESP8266, ESP32 series SoCs have dual core processor, more GPIO pins, faster Wi-Fi and BLE protocol support. The processor communicates with the peripherals through standard interfaces, i.e., UART, SPI and I2C. In normal (active) mode, ESP32 SoC requires 160-260 mA current to operate at 3.3 V DC supply voltage. When Wi-Fi and Bluetooth functions are open, the current drawn can spike

up to 790 mA. Although the ESP32 SoC is relatively power-hungry, the current drawn can be cut down by leveraging one of its sleep modes (modem sleep, light sleep, deep sleep and hibernation). Figure 9 shows the ESP32 SoC housed in QFN-49 package and its pinout diagram. ESP32 can be bought as a chip as well as module or development board forms, as shown in Figure 10. There are several variants of basic ESP32 SoC (ESP32- D0WD-V3, ESP32-D0WDQ6-V3, ESP32-D0WD, ESP32-D0WDQ6, ESP32-D2WD, ESP32-S0WD and ESP32-U4WDH) which have been introduced to meet the different needs of embedded system developers. Nevertheless, Espressif Systems has been upgraded to the ESP32 SoC in 2020. The latest ESP32 V3 SoC, known as ECO V3, offers a new silicon wafer as well as the secure boot and flash encryption. The ESP32- D0WD-V3, ESP32-D0WDQ6-V3 and ESP32-U4WDH SoCs are produced based on ECO V3 wafer. It is noteworthy that all of the ESP32 SoC variants share the same software development tools and they are largely code compatible. So, the firmware developed for one particular ESP32 SoC can be run on all others without significant modification. The specifications of the advanced ESP32 SoC versions can be summarized as follows:

• ESP32-S2 (2019, 43 programmable GPIOs) is a single-core Xtensa LX7 processor that has only Wi-Fi 4 (802.11 b/g/n) connectivity.

Figure 16



• ESP32-S3 (2020, 44 programmable GPIOs) has a dual-core XTensa LX7 processor with integrated Wi-Fi 4 and Bluetooth 5 (LE) connectivity. It has been designed to address the needs of the Artificial Intelligence of Things (AIoT) applications

• ESP32-C3 (2020, 22 programmable GPIOs) is a reduced version of the ESP32 and it has single-core RISC-V processor with Wi-Fi 4 and Bluetooth 5 (LE) connectivity.

STM32:

The STM32 F1xx micro-controllers are based upon the ARM CortexM3 core. The Cortex-M3 is also the basis for micro-controllers from a number of other manufacturers including TI, NXP, Toshiba, and Atmel. Sharing a common core means that the software development tools including compiler and debugger are common across a wide range of micro-controllers. The STM32 family

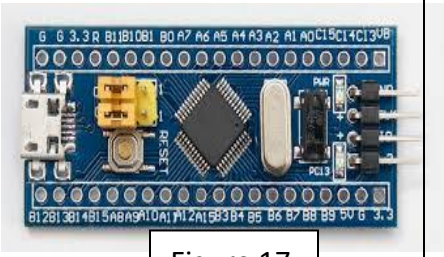


Figure 17

peripherals are completely different architecturally from the NXP family peripherals even where they have similar functionality

A block diagram of the STM32F100 processor used on the value line discovery board is illustrated in Figure x. The discovery processor has 8K bytes of SRAM and 128K bytes of flash. There are two peripheral communication buses – APB2 and APB1 supporting a wide variety of peripherals.STM32 F10x family. In the figure, I have included modules for general purposes I/O (GPIO), I2C, SPI, and serial IO (USART). Throughout this book I will introduce the modules as necessary.

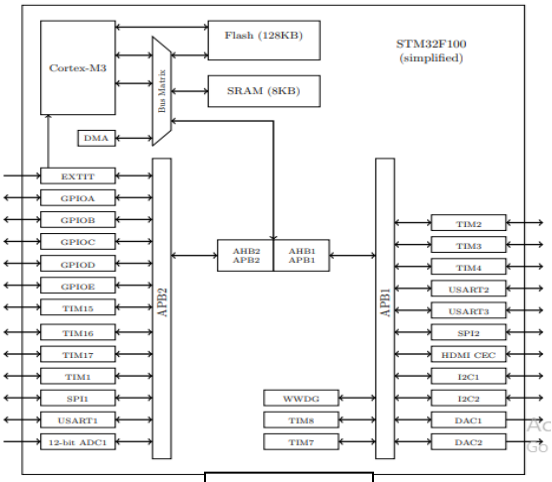


Figure 18

- **Sensors:**

A sensor is a device that detects and responds to input from the physical environment. Its main components include a receptor, transducer, signal conditioning electronics, and an output indicator. The receptor collects environmental input such as light, heat, sound, or magnetic fields. The transducer converts this input into an easily measurable form, usually an electrical signal. The signal conditioning circuitry amplifies or modifies the signal to match application requirements. Finally, the output indicator presents the result in a human-readable form, such as lights, sounds, or display readings.

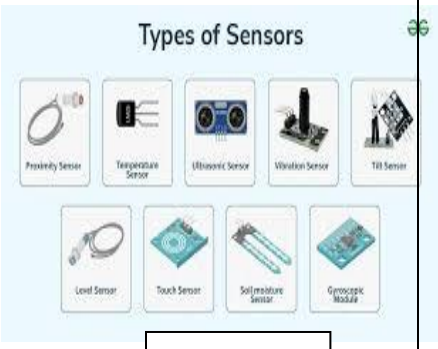


Figure 19

- **Function:**
  Sensors operate on the principle of taking an input and producing a related output. Steps include:

- Receptor: Detects environmental phenomena like temperature, light, motion, or magnetic fields.

- Transduction: Converts the input into another form of energy, e.g., thermal energy to electrical energy.
- Signal Conditioning: Amplifies and processes the signal, including filtering and digital conversion.
- Output: Provides a readable signal such as voltage, current, frequency, or digital code.
- Feedback (optional): Some sensors include feedback loops to improve accuracy or compensate for environmental changes.

• Types / Sensing Principles:

1. Mechanical: Detects deflection, stress, strain, or frequency/resistance changes to sense mechanical quantities.
2. Thermal: Uses temperature-dependent properties of materials (e.g., thermistors, thermocouples).
3. Electrical: Detects voltage, current, charge, or conductivity changes.

Examples:

Soil Moisture Sensor :

This Moisture Sensor uses Immersion Gold, which protects the nickel from oxidation. Electroless nickel immersion gold (ENIG) has several advantages over more conventional (and cheaper) surface plantings such as HASL (solder), including excellent surface planarity (particularly helpful for PCBs with large BGA packages), good oxidation resistance, and usability for untreated contact surfaces such as membrane switches and contact points.

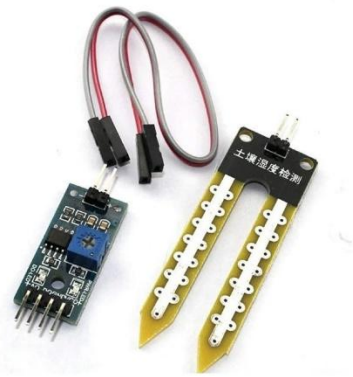


Figure 20

This Moisture Sensor can read the amount of moisture present in the soil surrounding it. It's a low-tech sensor, but ideal for monitoring an urban garden or your pet plant's water level. This Moisture Sensor can be used to detect the moisture of soil or judge if there is water around the sensor, let the plants in your garden can reach out for human help. They can be very easy to use, just insert it into the soil and then read it. This Moisture Sensor uses the two probes to pass current through the soil, and then it reads that resistance to get the moisture level. More water makes the soil conduct electricity more easily (less resistance), while dry soil conducts electricity poorly (more resistance).

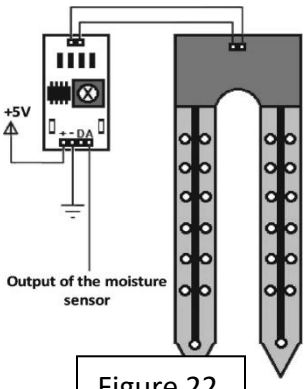


Figure 22

Water Level/Detection Sensor:

The Water Level/Detection Sensor is an easy-to-use, cost-effective water drop/level identification sensor that determines the water level by measuring the water droplets/water volume with a series of exposed parallel wire traces. It is easy to complete the conversion of water quantity to analog signal, and the output analog value can be directly read by the Arduino development board to achieve the effect of water level alarm.

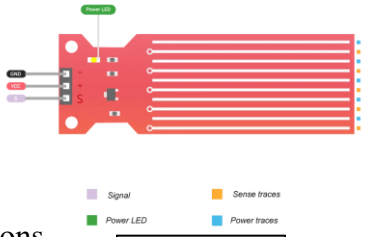


Figure 23

Features

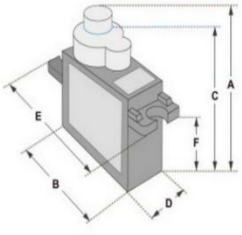
- High sensitivity to water presence and level changes.
- Analog output for continuous level monitoring.
- Simple interface with microcontrollers like Arduino.
- Low power consumption is suitable for battery-powered applications.

SG-90 Micro Servo Tower Pro 9g:

Interfacing an SG-90 servo motor is like a rite of passage in this craft. The Micro Servo Tower Pro 9g SG-90 is a lightweight servo motor that rotates 90 degrees in both directions.

Specifications:

- Operating Voltage: 4.8 ~ 6.0V
- Operating Speed: 0.12sec/60 degree(4.8V)~0.1sec/60 degree(6.0V)
- Torque: 1.8kg/cm(4.8V)
- Dead Band Width: 5usec
- Temperature Range: -30~60
- Cable Length: 25cm
- Brand name: Tower Pro
- location of each pin: red (+), brown (-), yellow (signal).



Dimensions & Specifications
A (mm) : 32
B (mm) : 23
C (mm) : 28.5
D (mm) : 12
E (mm) : 32
F (mm) : 19.5
Speed (sec) : 0.1
Torque (kg-cm) : 2.5
Weight (g) : 14.7
Voltage : 4.8 - 6



Figure 24

The Wire Mechanism

Here is the wire description you can consider to manage the circuit flawlessly:

- RED: Positive
- Brown: Negative
- Orange: Signal

Analog PH Meter Sensor

This Analog pH Meter is specifically designed for Arduino controllers, offering a simple, convenient, and practical solution for measuring water quality and other parameters. The kit is ideal for hobbyists and makers looking to add pH sensing to their projects without the complexity of more expensive meters.

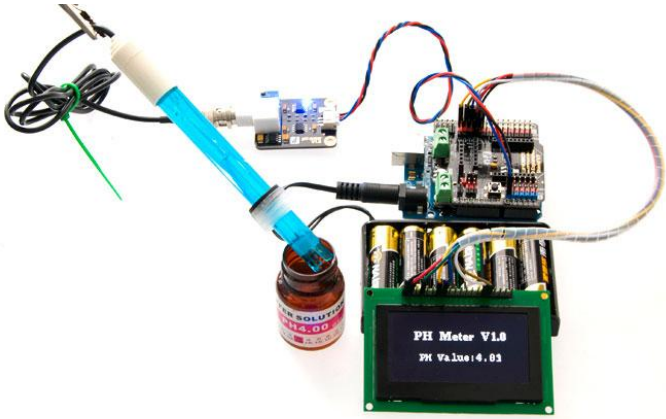


Figure 25

Features

- Specially designed for Arduino compatibility
- Simple and easy-to-use analog interface
- Onboard power indicator LED
- Standard BNC connector for reliable probe connection
- PH2.0 Gravity sensor interface
- Includes a pH probe with a 90cm cable

Specifications

- Measuring Temperature: 0 – 60 ℃
- Accuracy : ± 0.1pH (25 ℃ (at STP))
- Response Time : ≤ 1min

3- Fish Species (optimal conditions):

Fish Tilapia (Oreochromis niloticus):

Habitat & Water: Nile tilapia has been selected as the fish in our campus farm as they are tropical fishes and prefer warm water; 2530 C is the best place to keep them. The trick, however, is that they can in fact survive as low as 12 o C, and perform in a 5-10 pH range, and can therefore be used in both outdoors and indoors in Egyptian installations. In practical courses, such as a simulation of Cairo aquaponics, the pots typically have a



Figure 26

population of approximately 1,000 tilapia. They are also indigenous to Egypt hence they are very relaxed to slight changes in pH and temperature.

Diet/ Role: Tilapia are omnivores in nature. They consume algae, aquatic vegetation, insect and the traditional commercial feed pellets. Those feeds become protein rich and fast-growing muscle, and their droppings contain sufficient nutrients. Nitrifying bacteria change ammonia from the fish into nitrates which are digested by our plants. Tilapia provides us with the harvests both of fish and of organic fertilizer in a classroom or a small farm. They are fast, with no more than 600700g in 6-8 months.

Common Carp (Cyprinus carpio)

Habitat & Water: Habitat Common carp are the hardiest fish. They can survive, approximately, 14ºC to 36ºC therefore, they make a cool complement to tilapia during warm seasons and can also wait during cooler seasons. They are patients poor in water, and even low oxygen, less than 1mg/L is all alright with them. Middle-level hardness and a neutral pH that is neutral. Since they can live in the dark, cloudy water with high levels of nitrogen, we usually find them in outdoor ponds or rice paddies.



Figure 27

Food & Recent Habit: Carp are bottom predators. They graze on insects of water, worms, mollusca, and vegetation and rubbish. They are useful in an aquaponic environment, foraging on the residual feed and algae to prevent stagnation. They are also similar in what they eat to tilapia, but carp are not particular and will dig about a great deal. During the act of feeding and excretions, they force additional nutrients into the system. Temporary note: in certain markets no one values carp as a food and therefore at times it is disposed of as a nutrient cycle item and not a market product.

Prawns/ Shrimp/ Freshwater (Macrobrachium Rosenbergii).

Habitat and Water: These freshwater prawns or sometimes referred to as Shrimp require warm, clear water of a temperature of 26-30ºC. and an abundance of oxygen. Juveniles develop well in fresh waters although larvae require brackish waters (9-19p.l-1 salinity) in order to be successfully metamorphosed. This implies that a different stage of hatchery could be required during breeding. Within the classroom environment, we would probably start with freshwater juveniles in the post-larval stage.



Figure 28

Diet & Role: Prawn larvae simply bite planktons. Young and old are the ravenous scavengers-they will find food in zooplanktons, insects' larvae, plants and algae, and rotten plants or fish. The prawns would be useful in our aquaponic tank as they would go through the uneaten food and muck at the bottom cleaning up the waste products and supplying another source of fertilizers on their own.

Our aquaponic tank would benefit by having the prawns that would eat up the leftover feed and muck at the bottom cleaning up the accumulation of waste and providing another source of fertilizer. Sows of the prawn are female and give ton of eggs, which will breed in profusion under favorable warm and rich conditions.

Snails that dwell in water (e.g. Apple/Ramshorn Snails)

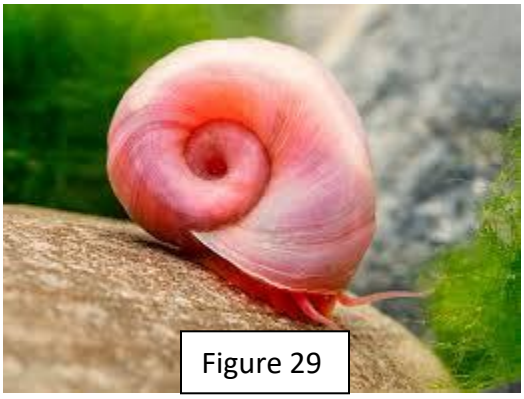


Figure 29

Habitat & Water: There are a lot of aquaponic snails, apple or ramshorn, which are amphibious. They possess gills and a lung hence they are not sensitive to low oxygen levels and can withhold temporary droughts. They like warm water (20-30°C) and are also sensitive to pesticides. Snails will flourish in an aquaponic tank provided the tank is in a stable environment.

Diet & Role: Snails are algae eaters that are scavengers. Marisa snails- large ramshorn type- are omnivores and they feed on decaying plants, algae as well as fish eggs. They browse biofilm and detritus cleaning up the tank. Basically, aquarium snails serve aquaponics its purposes: they decompose waste. The reproduction of one kind of snail may be rapid--in good conditions, with plenty of food, Marisa may in a couple of months become of an age when she is able to lay gelatinous egg masses on the surface of the water. We will have to control their population, but a little snail population can be rather useful.

Freshwater Crustacean (Crayfish /Crabs):



Figure 30

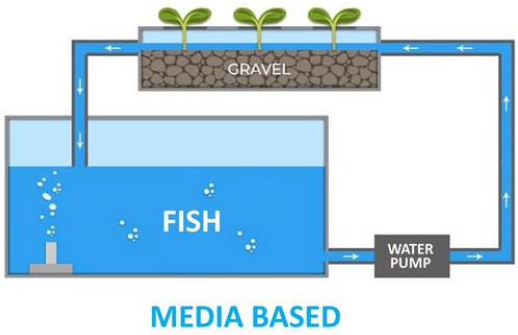
Habitat & Water: Crayfish are freshwater crawfish as we commonly refer to them. As an example, the Red Swamp Crayfish (Procambarus clarkii) will live comfortably between approximately 12 -30 o C. They are tolerant to low oxygen like living in water where dissolved oxygen levels are below 0.5mg/L; and like good hiding places, such as pipes or rocks. Bear in mind that crayfish are molar, therefore it is necessary to have some areas where they are able to shed off. A tight-fitting tank would be desired in our aquaponic system to ensure the animals do not burrow out, as well as a moderate level of filtration since these animals tend to stir up the sediment whenever they make any movement.

Diet & Role: The crayfish is an omnivorous detritivore and that means crayfish will eat rotting vegetation, algae, small invertebrates and fish remains. Their omnivorous nature of foraging is in fact useful to decompose everything that fish leave behind classified as organic. In a study of a pond, the red swamp crayfish consumed natural food resources such as algae and plants. They will scrape deposited feed and plant debris in a tank. They, however, do chew on the younger plants should they be hungry, hence the smart thing to do is to pick the appropriate plants and ensure primarily covering. Since they are crustaceans, they release waste that is composed of calcium and chitin thus providing the plants with a slightly varied nutrient composition.

4- Aquaponic System Designs:

Media Bed (Flood-and-Drain) System:

Figure 31



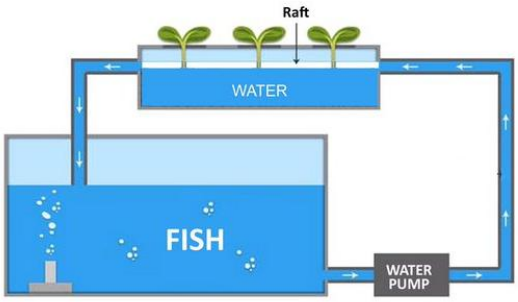
It consists simply of a container filled with inert material, such as gravel, expanded clay, rockwool or just whatever is available. into it we pour the water of a fish-tank, and either fill it up and empty it with a bell siphon or maintain a shallow circulation of some one inch. The media also makes the roots of plants contented and also removes waste materials of fish, which are left to a nitrifying bacterium to handle. The beds typically have a depth of 612 inches so that we may accommodate a number of crops. Due to the fact that the media entraps solids, we have to occasionally clean it or we can waste some worms by introducing them to aid the decomposition of waste.

Water Quality Requirement: Oxygenated water that is clean is desired by fish and bacteria in a media bed. Maintain dissolved oxygen (DO) at approximately 5mg/L or so and we will run air stones or simply bring the flow to stir the water up. Plants such as slightly acidic to neutral pH range which is typically 6.5-7.5. Quest on during cycling that range may be slightly longer (6.538.5) though as soon as it is within range we would make that range 6.87.0 as the ideal range in terms of nutrient uptake. The media is itself buffering a small opportunity of change in pH since it entraps solids, although we will watch the pH and hardness anyway. Water is circulated in the entire tank a number of times in a day; in the case of flood and drain bed; pumps circulate a few minutes at a time a few times per hour to flood and allow the media to be emptied and acquire oxygen.

System Operation: The media bed is a grow bed and a mechanical filter simultaneously. As floodwaters take place, the solid material is deposited at the surface and plants and worms decomposing the organics take place. It follows that we will have to carry out some maintenance, namely, siphoning off silt or shaking the media lightly in order to prevent anaerobic areas. Bell siphons (auto-drains) are usually prevalent, and they wash themselves when blocked. Electric pumps are not only required but can be operated with intervals to conserve any form of power. As the media has a tendency to dry during bad times between floods, we will construct a bed using a solid tub or a 55-gal IBC lined with growth media. It will also serve as a biofilter (bacteria hosting) and the gardening area. The arrangement is easy and closed system hence it is easy to install, though we have to wash the system and cycle it appropriately, and then before adding plants and fish.

Raft (Deep-Water System) System.

Design: A raft system (DWC) incorporates plants into floating rafts in a pond that has fish water of nutrient quality. The net pots are typically placed in boards or polystyrene rafts lined up along grow trays or channels- 6-12 inches deep. The water of fish tanks simply running beneath the raft, and therefore the roots are fully covered. The trays might be placed in troughs, or they can be at the ground level. They also operate long DWC channels in others such as Bustan Aquaponics in Egypt where lettuce and other greens are cultivated and fed by several giant fish tanks.



DEEP WATER CULTURE

Figure 32

Water Quality Requirements Rafts require a dense oxygen level since the roots never come out of water. We typically lose approximately a 1- in. air stone per 10 ft 2 of raft area. DO is recommended to remain at or above 6 to 1 L/L in order to have happy plants (that is higher than necessary by bacteria). Rafts are preferable in the pH of 6.8-7.0 and temperature of 20-30 o C (68-86 o F) like the media bed when we are dealing with tropical. Flow continuous pumps run round the clock to keep the water flowing and the roots nourished. We would like to achieve a minimum turnover of an hour. Owing to the constant contact between roots and water, any temperature or oxygen dip will strike the vegetation at unimaginable levels thus we will be even more vigilant in keeping a watch on temperature and DO. As an example, in Cairo location of Bustan they combine deep-water raft beds and greenhouse heating systems with tanks of tilapia to maintain a constant warm atmosphere.

Operation of the System: Raft systems are rather easy to operate- no media to clog to no flood-drain cycle to manage. However, the lord of aeration is oxygen, and we cannot afford to omit

aeration. This means that any solids that come up the stream are filtered by bacteria on the underside of the rafts, which become bio filters, then necessitating our pumps and filters to remove bio solid in the rafts. Cleaning them every now and then to say away the algae is necessary, however, picking them is a walk in the park. Plants in net pots on the raft will be stuck in the net pots, a pump (and perhaps a small clarifier) used to keep the water circulating and the raft continually flooded so as to keep the roots constantly immersed in nutrients--just be sure to keep the oxygen level up. The levels of nutrients and the pH will remain consistent as per the media bed system and therefore we will have one common water-quality target.

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5. Techniques for plastic recycling:

Introduction of recycling

Plastics are one of the world's most-used materials. As its name suggests – from the Greek meaning capable of being shaped or molded – plastic can adopt any shape or form. This is why it is used for such a wide variety of applications, from everyday single-use products like packaging and bottles to products that last for years, such as furniture, clothes, building materials and automotive components. Plastics have replaced a wide range of traditional materials including glass, steel, wood, and even concrete. Plastics weigh less, cost less and off er outstanding technical properties. The rise of plastic coincides with the years of post-World War II prosperity and the burgeoning consumer society. French intellectual Roland Barthes celebrated it in his 1957 book Mythologies. "So, more than a substance, plastic is the very idea of its infinite." He also made this prophecy: "The hierarchy of substances is abolished, and a single one replaces them all – the whole world can be plasticized and even life itself since, we are told, they are beginning to make plastic aortas." The amount of plastic used has indeed grown constantly over the past 30 years, reaching over 300 million metrics.

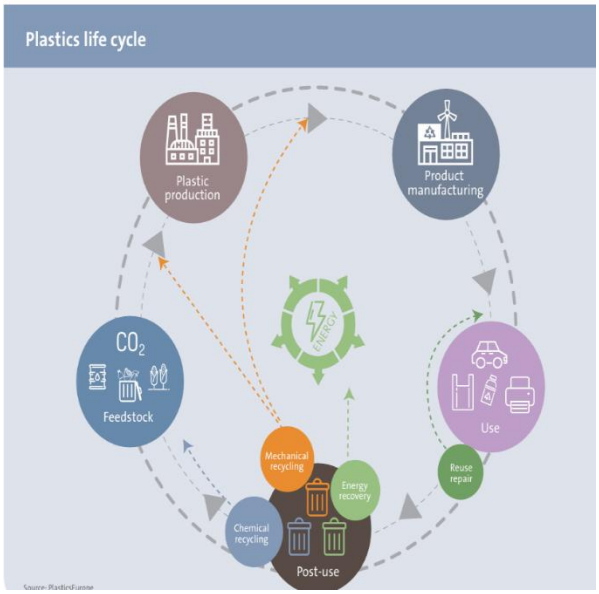


Figure 33

Plastic waste management

Over 90% of raw plastic is produced from fossil fuels (oil or natural gas). The polymers are synthesized by major petrochemical companies like ExxonMobil, Sinopec and Total. The plastic is then sold to plastic manufacturers to make objects, mostly by injection, blow molding or heat forming. These objects are then assembled or sold directly by brand owners via a range of retail circuits.

Plastic objects become waste once products are consumed, and waste management is extremely variable from country to country. There are four broad groups of countries:

- developed economies with regulations that encourage recycling, and developed economies that do not have such incentives.
- developing economies with large industrial bases and developing economies with little industrial activity.
- Developed economies with regulations that encourage recycling tend to be mature economies (rich country, modest growth) with good traditional waste management infrastructure (landfill, energy recovery) and relatively high labor costs.
- This applies to Western Europe and Japan. Regulations to encourage recycling come in a variety of forms. It is quite common to set up organizations to oversee recycling. These organizations are used to finance some of the costs involved in collecting and sorting plastic waste. Funding generally comes from suppliers – producers and retailers – or is raised from consumers via green levies. This allows externalities relating to end-of-life management to be re-internalized into product pricing.
- Recycling in these situations relies on significant infrastructure for sorting and processing plastic waste by polymer type, capable of producing recycled plastic suitable for reuse by manufacturers. These countries also use measures to increase the cost of traditional processing solutions, in the form of taxes on landfill and incineration. Countries in this category can attain recycling rates in the order of 30%.
- Developed economies without regulatory incitement possess characteristics similar to the first group, but they focus on traditional waste management methods: landfill and incineration. These countries are typified by the USA and Australia. Recycling remains underdeveloped and marginal in the absence of specific regulations to boost its competitiveness relative to other forms of processing. Less than 10% of plastic waste is recycled locally

Where are the plastic recycled

Most waste is recycled locally, either in the producer country or a nearby country, but a sizeable export industry has also emerged over the past 30 years. This market essentially involves flows to

China, where the material is in high demand, from developed economies (both those with and without regulations to incite recycling). This waste export market takes advantage of low freight rates for return legs on bulk container carriers after they have offloaded cargo from China at ports in Europe and the United States.

In 2017, Europe exported over 2 billion metric tons of plastic waste to China. This market is currently in a transitory period because of the Chinese government's ban on the import of post-consumer waste that came into force in January 2018. New markets have emerged via Southeast Asia, but, likely, these countries will also ban imports. These changes represent a major challenge to recyclers as they concern very large volumes. However, the long-term effect of such measures is to encourage local recycling.

What happens to the recycled plastic

Recycled resins can deliver attractive technical properties and are suitable substitutes in many applications. There are pretty much as many possible uses for it as there are for raw plastic: bottles, fabrics, packaging, automotive, household appliances, construction, etc. Recycled plastics meet around 10% of global demand for plastic.

Some applications are harder to service, however, for technical or regulatory reasons. For example, food-grade certification requires a feedstock with very high levels of traceability. It is also difficult to substitute 100% recycled resins for certain plastics able to withstand very high pressures. Bear in mind that, historically, users of plastics would buy recycled plastic because it was cheaper than virgin plastic.

We must accelerate the recycling of plastic

Currently, most plastic waste goes to landfills or is released into the environment one way or another. Every year, in Southeast Asia and China, 4 to 12 million metric tons of plastic packaging are swept down rivers and end up in the oceans. This plastic takes hundreds of years to decompose and constitutes a grave threat to the marine environment.

Recycling plastics also leads to significant reductions in atmospheric emissions of CO_2, because using recycled plastic avoids the emission of an amount equivalent to that generated during the production of raw plastic

What are the advantages and benefits of recycling

Developing recycling also fosters local growth by re-internalizing employment within a territory. Typically, a plant producing about 50,000 metric tons of recycled plastic will employ around 30 people. This is significantly more jobs than those generated by sending an equivalent amount of waste to landfill or incinerating it, or by the petrochemical industry synthesizing an equivalent quantity of virgin resins – and these jobs are local.

Setting up a system to recycle plastic waste allows local industry to emerge and recover value from the recycled material. Where there is no recycling, energy recovery is the only income-generating possibility.

However, because plastic waste recycling systems are logistically more complex than traditional waste processing systems (separate collections, differentiated flows, etc.), this leads to higher waste management costs. This additional cost has to be covered by producers and consumers of plastic goods through extended producer responsibility (EPR).

Developing this activity also helps to deliver resource independence to countries with few oil or gas resources, because making raw plastic requires crude oil or natural gas.

IV- Prior Solutions:

Vertical farms:

They are a type of Controlled Environment Agriculture (CEA). CEA strategies are far from new, as they emerged in the 1700s. They have traditionally taken the form of greenhouses, which have greatly contributed to the global food supply over the last decades. Vertical farms diff er from greenhouses in terms of their height. Indeed, a vertical farm can be simply perceived as a stack of greenhouses on top of each other. Therefore, for the same amount of ground space used, the plant yield is multiplied by the number of floors of the vertical farm. The higher the vertical farm, the more produce it yields, albeit with the same footprint. Consequently, they are now capable of producing millions of tons every year. Vertical farms mainly diff er amongst each other in terms of the technological methods used to grow edible plants indoors.



Figure 34

• (1) **Hydroponics:**

Hydroponics has been used successfully to grow a variety of crops such as lettuce, tomato, cucumber, herbs and many types of flowers . Its advantages over conventional production systems are faster growth, higher productivity, easier handling, greater water efficiency and lesser use of fertilizers. In hydroponics, the concentration of nutrients can be controlled in the aqueous solution making it easier to observe the symptoms of nutrient deficiency or toxicity in.

Hydroponics has proven to be useful in many fields such as in toxicological studies on the accumulation and destination of emergent contaminants in plants, the implementation of native and exotic crops with experimental, commercial or medical purposes, as well as the cultivation of traditional crops like vegetables and ornamental plants.

In recent years it has been demonstrated that hydroponics can be used as a method for the treatment of wastewater and thus, reduce the pollutants load to the environment. Due to its high nutrient content, reclaimed water (RW or treated wastewater) can be a source of water for hydroponic systems. By combining RW with hydroponic production, it is possible to obtain safe and viable agricultural produce, and to provide additional treatment to RW prior to its discharge to the environment. This is of special importance in arid regions of the world, where freshwater resources are rapidly being depleted and wastewater is discharged into the ocean and other water bodies without any further social or environmental benefit.

The present review includes the history of hydroponics, its different uses, advantages and disadvantages and a discussion about the future of hydroponics as a method for organic produce cultivation using RW, including the new market for legal cannabis. The goal is to look into the advantages of hydroponics and evaluate the success of using RW as a nutrient solution. Hydroponic systems here are defined as soil-less systems used for the production of vegetables, fruits and ornamental flowers.

Types of the circulation systems in hydroponics:

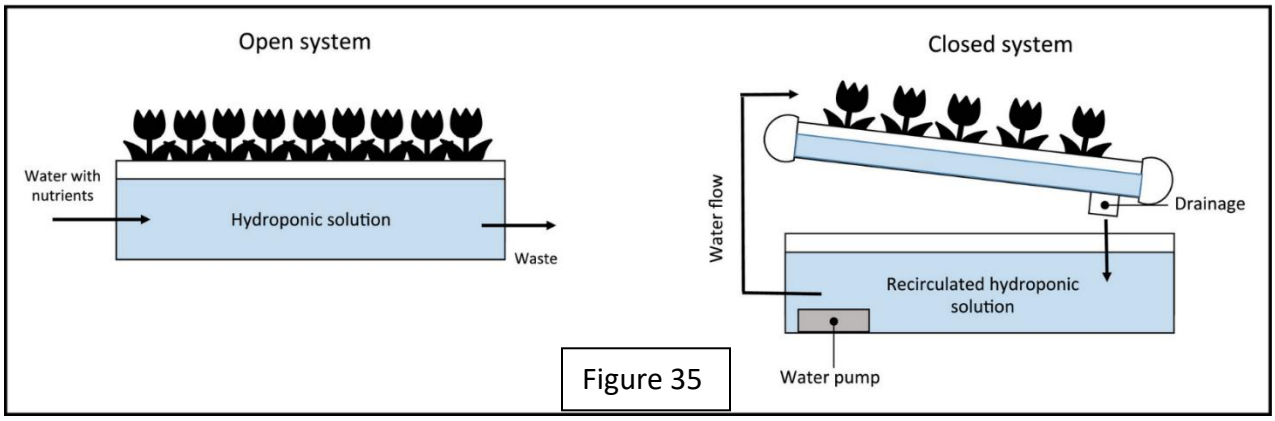


Figure 35

Open systems:

In open systems, the roots are in contact with the nutrient solution either permanently or occasionally. The nutrient solution and the media substrate are used only once, meaning that it is not recirculated or recycled. Some advantages of the open systems are the simpler application of the nutrient solution and the reduction of the risk of infection on plants.

Closed systems:

In the closed system, the nutrient solution is applied to the roots of the plants, collected in a container and returned periodically to the crop (most of the hydroponic systems fall into this group). Plants grow either in liquid solution or on a solid substrate. These substrates can be of organic nature (such as sawdust, rice husk, coconut fibre and charcoal) or inorganic (sand, gravel, pumice or ground brick). Water and nutrients are used in an optimal manner yet the main disadvantage is the dependency on electricity.

Types of hydroponics and soilless systems:

Some types of hydroponic systems commonly used are wick or passive system, deep-water culture, nutrient film technique and media bed based.

- **Wick or passive system:**

The wick or passive system is considered to be a less expensive technique in which there is no recycling of the nutrient solution but rather the plants absorb the solution through a system that uses the capillary capacity of the roots and fibres that transport the water to the plants. This system is used mainly in small-scale production and is not recommended for crops with a long-life cycle and can be used with open or closed circulation.



Figure 36

- **Deep-water culture:**

The technique of the deep-water culture system was the precursor to most of the systems currently used. It consists of an aeration and pumping system and a platform in which the crops grow. The continuous aeration allows for the roots to be submerged within the nutrient solution. The oxygen concentration, conductivity and pH have to be controlled to optimize plant growth. This type of system can be used with open or closed circulation.



Figure 37

- **Media bed based (Ebb & Flow):**

This system uses a tank or container filled with substrate. The crops are flooded with the nutrient solution and then drained periodically. This system requires constant supervision as a malfunction could cause the dehydration of the plants in a short period of time. A limitation is the rapid dissemination of bacterial or fungal diseases on the roots.

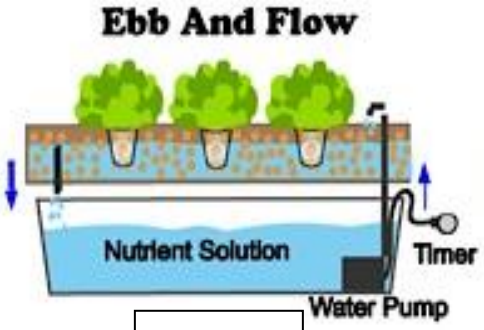


Figure 38

- **Nutrient film technique:**

One of the most common techniques in closed hydroponic systems is the Nutrient Film Technique (NFT), developed by Allan Cooper in 1960. In this technique, a thin layer with the nutrient solution is provided to the roots, which also allows for the necessary oxygen. Originally the plants were grown on rock wool in containers made of opaque polyethylene. Currently, the NFT systems have been modified and a variety of supporting media for plant growth are used. This type of system can be used with open or closed circulation though the latter is more common.

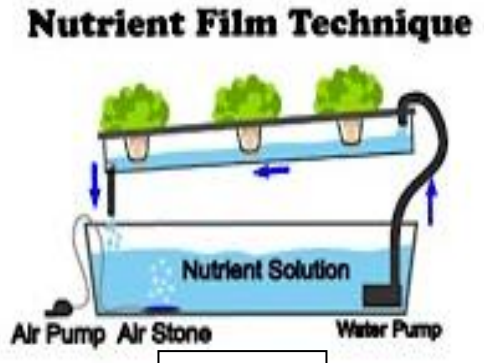


Figure 39

• (2) Aeroponics:

Aeroponics is the science of plant cultivation without incorporation of the soil or a substrate culture. Where plant grows in the air with the assistance of an artificial support and no soil or substrate is required to support the plant. Basically, it is an air water culture cultivation system, the roots of plant are hanged inside a sealed container under darkness and openly exposed in the air to get water nutrient-rich spray through atomizers. The upper portion of the plant leaves and crown extend above the wet zone. The root and canopy of the plant are separated by the artificially provided structure. The system uses the nutrient-enriched spray in the air with the help of pressure nozzles or foggers to sustain hyper growth under controlled.

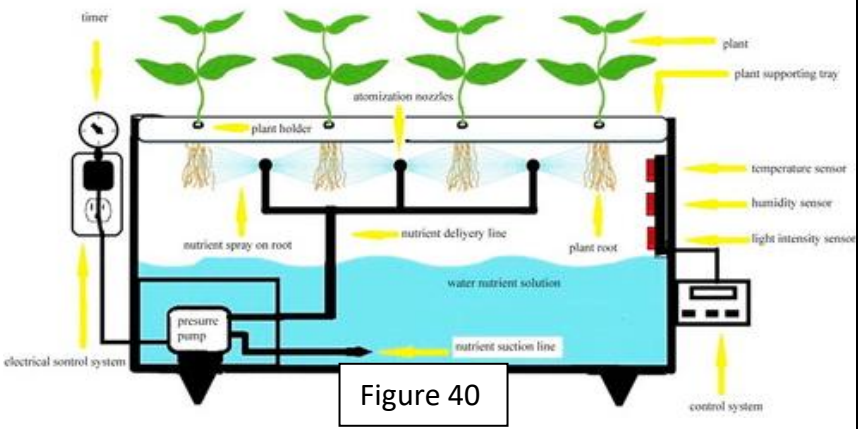


Figure 40

In The air-growing culture reduced the mechanical injuries and interferences with significant growth compared with soil, sand, or even aerated water culture. The discrete nature of atomization spray interval and duration allows for measuring the nutrient uptake concentrations within the plant over time under different conditions. The atomization spray provides the intermittent mist of nutrients to plant root on different periodic intervals for the specific duration rather than a permanent misting. Only aeroponics system is the most efficient plant cultivation system to grow the plant without the interference of soil comparing with other soil-less techniques. The nutrient-mist system uses a minimum amount of water and provides an excellent environment for plant growth.

Case Study:

Arbuscular mycorrhizal fungi production and plant cultivar differences in root growth. The system achieved the performance for saving the water up to 99%, nutrients 50% and 45% less time than soil-based cultivation.

The controlled conditions include uniform nutrients concentration, EC and pH values, temperature, humidity, light intensity, atomization frequency, atomization spray time, atomization interval time, and oxygen availability. However, the plant grows speedily in the system due to the sterile environment and abundant oxygen availability in the growth chamber. However, several research studies practiced the modern plant growing technology for the cultivation of horticultural ornamental, the root of herbs and root based medicinal.

• (3) Hybrid aquaponics,

Aquaponics fits into the broader definition of integrated agri-aquaculture systems (IAAS). However, IAAS applies many different aquatic animal and plant production technologies in many contexts, whereas aquaponics is far more tightly associated with integrating tank-based fish culture technologies (e.g. recirculating aquaculture systems; RAS) with aquatic or hydroponic plant culture technologies. RAS technologies apply conserved and standard methods for the culture of fish in tanks with applied filtration to control and alter the water chemistry to make it suitable for fish (i.e. fast and efficient solid fish wastes removal, efficient, bacteria-mediated conversion of potentially toxic dissolved fish waste ammonia to less toxic nitrate and oxygen maintenance via assisted aeration or directly injected oxygen gas)

Whilst aquaponic systems do integrate tank-based aquaculture technologies with hydroponic plant culture technologies, aquaponic systems work by supplying nutrients to, and partitioning nutrients between, the production inhabitants (fish and plants) and the inhabitants that perform biological and chemical services that assist the production inhabitant outcome (microflora) (Fig. 5.1) .

Schematic representation of the nutrient flows within an aquaponic system. Fish feed is the major nutrient entry point. The fish eat the feed, use what nutrients they need, release the rest as waste and this waste is then partitioned between the microbes, plants and system water.

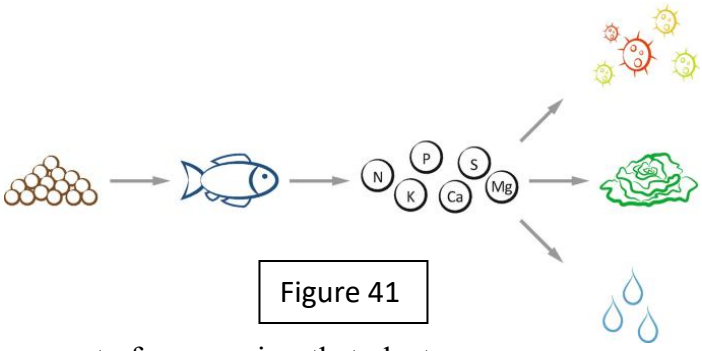


Figure 41

It must also be stated that the proportion of fish to plants should remain at a level that supports a core prospect of aquaponics; that plants are grown using fish wastes.

The name aquaponics, until relatively recently (i.e. the last 3–5 years), has been universally applied to coupled and fully recirculating system designs that seek to supply as much of the required plant nutrition from the fish wastes as possible.

Simplified scheme of the main water flows within a coupled aquaponic system. The nutrient concentrations in the process water are equally distributed throughout the whole system

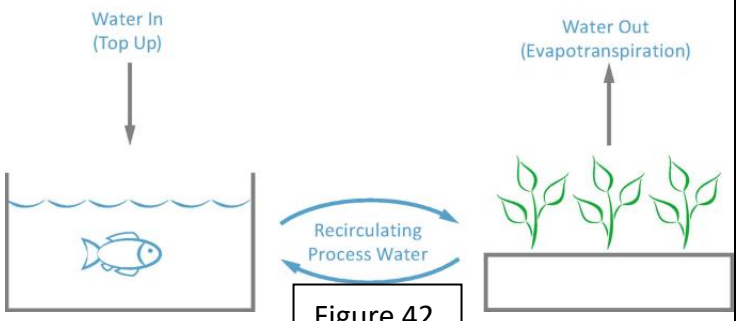


Figure 42

However, decoupled approaches now represent a proportion of the systems being researched or commercially applied, especially in Europe, and in current practice do not supply plant nutrient requirements from the fish wastes to the same extent as fully recirculating systems do state that for 3 examples of current European decoupled aquaponic system designs, the relative addition requirements for external hydroponic-derived nutrients are 35–60%. Because these decoupled designs are based on integrating existing RAS and hydroponic/substrate culture technologies, they are regarded as aquaponic in nature.

Simplified scheme of the main water flows within a decoupled aquaponic system. The nutrient concentrations in each component may be separately tailored to the individual component requirement.

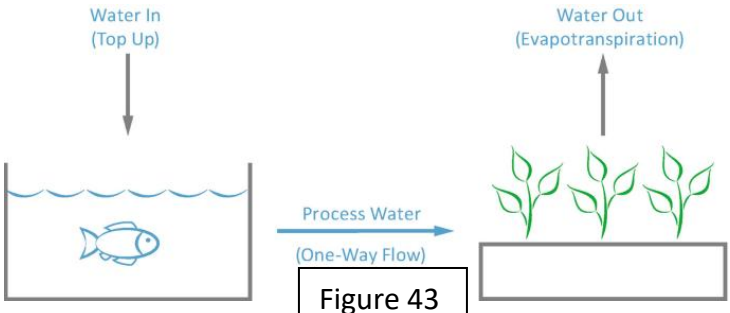


Figure 43

Applications:

Techno Farm (Japan):



Figure 44

Its new Techno Farm, expected to open as early as this month, will push efficiency further, yielding 648 heads of lettuce a square meter annually, compared with 300 heads at its Kameoka farm and only 5 in an outdoor farm. It will use only 110 milliliters of water as a lettuce, 1 percent of the volume needed outdoors as moisture emitted by the vegetable is condensed and reused.

Power consumption per head will also decrease, with the new factory using custom-designed LEDs that require about 30 percent less energy. A collaboration with telecoms company NTT West on an artificial intelligence program to analyze production data could boost yields even more.

Spread does not disclose the cost of producing lettuce at its farms, but Japanese researcher Innoplex estimates the cost to produce one head of lettuce at its existing Kameoka building is about ¥80 — among the lowest in the world.

But extreme weather events and climate change, major disruptors of traditional agriculture, are making vertical farming competitive even sooner. Japan's hottest-ever summer this year, with its heavy rains, typhoons and flooding, sent supermarket lettuce prices soaring to more than double the level at which Spread sells its products.

Hybrid Aquaponics

Nelson and Pade Aquaponics Farm (USA):



Figure 45

Nelson and Pade is a company that designs and builds aquaponics farming systems that grow fish and plants together in one system. The company started in 1985 as a hydroponics business, then later developed aquaponics systems based on scientific research. The kinds that they use. Fish (catfish, perch, bluegill, trout) plants (lettuce, basil, mint, tomato, peppers, spinach). The system grows vegetables, raise fish in tanks, reuse water repeatedly, and produce food all year.

Merits:

-Safety Factor: The system of the farm Is working efficiently and safe for food production.

The parts that are used in the farm

- Bio filter
- Water pumps
- Plastic fish and plant tank
- And this can be like our project that can recycle this plastic

Demerits:

-**Expensive**: Sufficient funding is necessary for this contraption to be properly executed.

-**High Energy consumption**: The system consumes enormous wattage which in turn restrains the design from being implemented within regions that lack the privilege of a nearby powerplant (arid areas).

-**Professional Labor required**: Although the system is semi-automatic, it requires

Professional human intervention to keep everything running smoothly.

Bustan aquaponics

Bustan aquaponics in the first farm in Egypt, created in 2011. It produces fish and vegetables using a suitable farming system that combines aquaculture and hydroponics. Who made it is Faris Farrag. Egyptian and former investment banker. He started the farm after returning to Egypt in 2011 with the goal of improving food sustainability and water efficiency.

The farm focuses on:

1. Suitable agriculture
2. Water transfer
3. Local food supply for Cairo

Farm size

Around 1,000 square meters. The aquaponic system produces high yields compared to traditional farms.

Production System (operating mechanism):

The farm uses aquaponics, combining fish are grown in specialized tanks, and plant production.



Figure 46

Natural Cycle

1. Fish produce waste.
2. Bacteria convert waste into nutrients.
3. Plants absorb nutrients.
4. Clean water returns to the fish tanks.

This creates a closed-loop ecosystem. This is very important in Egypt where water resources are limited.

Products

The farm produces both fish and vegetables (hence the name aquaponics).

- **Fish:** Nile Tilapia (main fish species) with **Annual gross production** of Around 30–35 tons per year.
- **Vegetables and Greens: Examples grown at the farm:** Lettuce (several varieties) Spinach with an **Annual vegetable production of** Around 80 tons of leafy greens per year.

Merits:

Saves Water: Aquaponics systems like those used at Bustan Aquaponics recycle water continuously. it uses about 90–95% less water than traditional farming. This is very useful in dry countries like Egypt.

Produces Two Foods at the Same Time:

The farm produces:

- Fish such as Nile Tilapia
- Vegetables like Lettuce, Basil, and Spinach

So, a single system produces both protein and vegetables.

Absence of Chemical Fertilizers:

Plants get nutrients from fish waste

This makes the food more natural and safer.

Environmentally Friendly

The system provides less water usage, less overall pollution, and recycles nutrients naturally, which in return make it suitable for sustainable agriculture.

Demerits:

High Start-Up Cost:

Building an aquaponics farm requires fish tanks, water pumps, a reliable pipeline system, Filters, and Greenhouses for all year production, making the initial investment expensive.

Necessity of Technical Knowledge:

Farmers must monitor Water pH, oxygen levels, Temperature, and Ammonia levels

If these are not controlled, fish and plants could be potentially prone to perishing.

Dependence on Electricity:

The system uses pumps to move water. If electricity stops:

- Water circulation stops
- Fish may die quickly.

System Balance Is Sensitive:

Aquaponics depend on the balance between:

- fish
- bacteria
- plants

Gaza Rooftop Aquaponics Pilot

One of the preceding hybrid aquaponics systems was the Gaza City rooftop garden project (2012 13). Families participating in the study were provided with 1m 3 fish tank attached to a grow-bed a 4m 2 (flood-and-drain system) of gravel. This closed loop system cultured tilapia and vegetables simultaneously: water which was enriched by fish waste was pumped out of the tank and into the plants where it got filtered into. During a time period of one year a well stocked unit generated about 20-25 heads of lettuce every week and 35-40kg of fish. Such outcomes (around 360 lettuce heads and about 3040 kg fish per annum per unit) were recorded to happen in actual project accounts. Majority of the families were able to produce vegetables and fish and therefore proved the efficiency of the system.



Figure 47

System Design and Operation

Parts: A fish tank made of fiberglass (tilapia), a grow-blood (volcanic gravel media), a submersible pump, and primitive filters.
Operation: Pumping: The pump will pump out and pump in fish-tank water through the grow-bed. Ammonia is converted to nitrates by bacteria, and its absorption by plants goes back to the fish in clean water. Another possible fertilizer to be added every 1–2 weeks is worms or biosolids.
Management: The required operational knowledge was gained by the participants within a short period of time. FAO workers educated the families on how to feed livestock, control water PH and regulate the flow. The system would need minimal labour, which was mainly fish feeding and some plant pruning.

Outcomes and Impact

Crops: People said that there was plenty of summer vegetables (tomatoes, peppers, lettuce). On a pilot assessment, 25 lettuce heads per unit were reported to be harvested weekly.
Fish Production: At the end of the year, a single unit was able to produce around 3040 kg of tilapia, which is incredibly abundant in protein to the house.

Food Security: most of the families used their food production and fish thus buying less external food. The respondents talked about the system as being very helpful. No family gave up on the study and everyone went ahead using the kit.

Efficiency in Water and Resources.

An example of water saving capabilities of aquaponics is the Gaza systems. Due to the recirculation of the water, the consumption is low as compared to traditional soil farming. This was the advantage that was pointed out by the FAO project, but no distinctive chart was provided. On the FAO reports, 4m2 of aquaponic unit uses about 10 per cent of the water that a garden based on soil would have used.

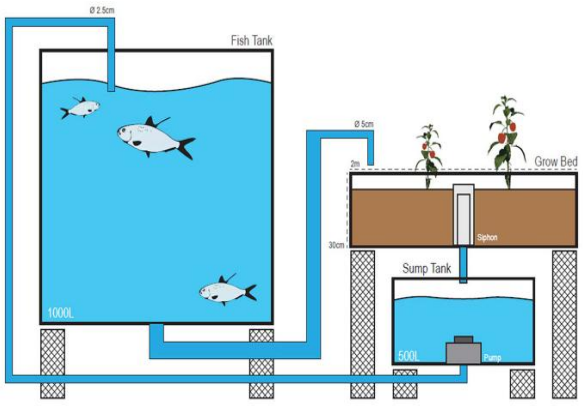


Figure 48

Nutrient Reuse: The process of nutrient reuse is very efficient; the nutrient waste by fish alone was sufficient to nourish the vegetables eliminating the necessity of having additional chemical manure.

Chapter II

(Generating and Defending a Solution)

I. Design Requirements
II. Selection of Solution
III. Selection of Prototype

I. Design requirements:

In order to successfully generate a suitable solution, the following requirements must be met. These requirements are:

- Prototype Scale and Functionality:

The prototype must be a small-scale, functional unit-either a maximum 10 L aquaculture system or a 0.25 m^2 crop growing area-capable of operating continuously for at least 6 hours without manual intervention. The prototype must be portable, testable, and fully operational on the exhibition day.

- Construction Materials:

The prototype must be constructed using waste or recycled materials.

- Controlled Variables and Setpoints:

The team must define at least three environmental parameters, with only one allowed to be temperature, pH, or salinity. For each parameter, a target operating range (setpoint window) must be established based on the selected fish species (aquaculture), plant species (greenhouse agriculture), or both (hybrid aquaponics). The selected ranges must be appropriate to the school region's environmental and agricultural needs.

- Sensor Selection and Calibration:

The system must include appropriate sensor types corresponding to the selected parameters. All sensors must be properly calibrated using standard calibration methods. Calibration data, procedures, and calibration curves must be documented and included in the Materials and Methods section of the project poster.

- Actuator Implementation:

The system must include at least two functional actuators capable of regulating variations in the three selected parameters. Actuators must operate in response to sensor feedback through the control algorithm. The power consumption (in watts) of each actuator must be measured and reported in the Results section.

- Control Algorithm Design:

At least one control strategy must be implemented, such as on/off control, rule-based expert logic (IF–THEN rules), or adaptive or scheduled control algorithms.

- Variation from Normal Conditions and Alert Testing:

Identification of at least three realistic deviations from normal conditions for each of the three selected parameters should take place, and demonstrate how the control system detects, alerts, and tolerates these variations. Additionally, students must record the time required to restore system stability after each deviation.

- Data Logging and Real-Time Monitoring:

The team must record three readings for each of the three selected parameters at fixed time intervals throughout the day for five consecutive days outdoors. All collected data must be documented in the project portfolio and presented on the poster. If any parameter falls outside its normal operating range, the response time required to restore normal conditions must be recorded.

- Database Library:

The prototype control system must include a preloaded database containing environmental condition ranges for at least two different plant species or aquatic organisms, based on the selected feedback control system.

- Data Visualization:

The team must provide graphs illustrating each parameter over time. Where appropriate, graphs should include error bars or confidence intervals.

- Prototype Functionality and Submission:

The prototype must be functional, testable, and portable on the exhibition day.

- Natural Materials Preference:

The body of your prototype (except electronic components) must be constructed using recycled or waste materials.

- Portability and Testability:

The prototype must be testable at any time and in any place.

- Cost-effectiveness:

The prototype cost will be evaluated as part of the assessment.

- Ethical Code of Conduct:

Humane measures should be followed when dealing with living organisms included in your system.

II. Selection of Solution

Our capstone idea involves the production of a means to elevate the gross annual produce of either indigenous/domestic crops, Egyptian aquaculture, or both simultaneously. There were many ways to reach this as discussed in Research with its certain advantages and disadvantages and with each concept needing specific environmental requirements.

After looking at multiple points of view on the problem and the challenge of our capstone, and after searching and analyzing many different Prior Solutions, pursuing the current idea of our capstone was the most viable option, which is the combination of both aquaculture and hydroponics using a hybrid aquaponic system (HAS) utilizing the work of mother nature combined with some intellectual engineering. That will happen due to the fact that fish excrement contains a high level of ammonia, which in turn becomes highly absorbable nitrogen for the chosen plant, filtering the water in the process.

The construction of a hybrid aquaponic system has its own set of merits and demerits in many ways.

Advantages:

- **Eco-friendly Production:**

- HASs do not involve the use of harmful fertilizers; however, it runs on fish excrement as mentioned above, which in turn helps restore the environment.
- It's renewable, clean, and gets rid of waste while using nature's power and basic chemical processes to create valuable products that would be used in many useful ways.
- Does not have a carbon footprint nor produce any harmful greenhouse gases, as the CO_2 produced by the aquatic creatures circulates.

- **Suitable in Diverse Environments**

- As the system involves both aquaculture and agricultural subsystems, a lot of variations could be implemented to further suit the surrounding environment (e.g., swapping the aquaponic system with an aeroponic system for hotter climates).
- Hydropower tanks are used in the generation of eco-friendly electricity without emissions.
- Choosing fish like Tilapia or Catfish could enable them to thrive in harsh conditions.

- **High potential operations:**

- They can be implemented in small-scale projects like a small domestic contraption without the need for huge infrastructures that would be costly and waste a lot of funds and energy in the construction process.
- They can be implemented on large scales over lakes, roofs, waterfalls, or even streets and buildings, producing steady, high-quality produce suitable for consumption.

To conclude, HASs, which consider the flourishing of both aquaculture and hydroponics via chemical reaction and biological mutualization, are excellent at producing a steady, high-quality produce suitable for consumption, using sensitive machines and operations providing safety, sustainability and a good alternative source of aquatic culture and crops simultaneously that can be flexible suited for the surrounding environment and without the limitations of having a strong infrastructure to operate.

Disadvantages:

- **High operating standards**

- Unlike other means, the HAS requires operation under the supervision of professional cadres, even if the system runs automatically. This is due to the involvement of Acid-Base titration, Ammonia treatment, and other parameter management and actuator evaluation.

- **Relatively High startup cost:**

- **The Nelson and Pade aquaponics system cost relatively about $112,000 for large-scale commercial setups**, which is estimated to be approximately **6,000,000 EGP**.

III. Selection of Prototype:

Providing the most optimal/suitable environment for both the chosen crop (Mint leaves) and the aquatic creature (Indigenous Nile Prawns) is the desired objective. This should only be achieved via a sophisticated contraption to monitor the selected parameters (pH, Turbidity, Tank Water level, Soil Moisture) that are tailored specifically for the chosen organisms, and if a shift in the surrounding environment takes place and the sensors are stimulated, the system is to respond, activating the actuators to oppose the change in the process.

The Prototype of the project consists of 2 subsystems. These systems are:

The Feedback Provider:

It includes the following Sensors:



Figure 49

- **Soil Moisture Sensor:**

This Moisture Sensor uses Immersion Gold, which protects the nickel from oxidation. Electroless nickel immersion gold (ENIG) has several advantages over more conventional,

Merits:

- Durability: It has oxidation resistance properties, which help increase its lifetime.
- Flexibility: Usable for a wide range of untreated contact surfaces such as membrane switches and contact points.

- **Water Level/Detection Sensor:**

The Water Level/Detection Sensor is a cost-effective water drop/level identification sensor that determines the water level by measuring the water droplets/water volume with a series of exposed parallel wire traces. It is easy to complete the conversion of water quantity to analog signal.



Figure 50

Merits:

- Accuracy: It is highly sensitive to water presence and level changes.
- Continuity: It provides analog output for continuous level monitoring.
- Conservative Power Usage: it possesses low power consumption, which is suitable for battery-powered applications.

Analog PH Meter Sensor

This Analog pH Meter is specifically designed for convenientcy and utter practicalillity, designed for measuring water quality and other parameters.



Figure 51

Merits:

- Accuracy : It involves a measurement error of ± 0.1pH (25 ℃ (at STP)).
- Swiftness: It provide an accurate result every 60 seconds.
- Reliability: Due to the accuracy it provides, they measurements it generates are quite reliable.

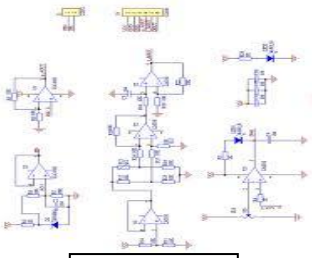


Figure 52

- **TS-300B Turbidity Sensor Module:**

an optical device used to measure the cloudiness or haziness of a liquid caused by suspended particles, as these particles could pose a threat to both the fish and the crops.



Figure 53

The Action Subsystem:

It includes the following actuators:

- **Alkaline/Acid:**

They will be put in glass vials that are connected to the main fish tank in order to stablize the pH if the pH sensor indicates a high level of pH. The servo motor will be used as a gateway for both of them.

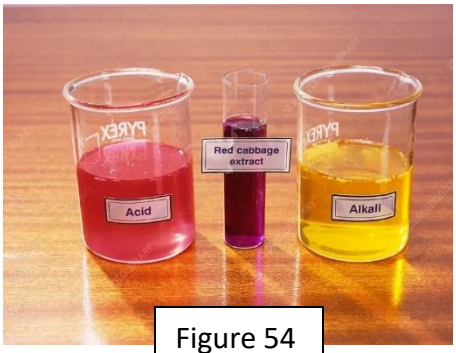


Figure 54

- **Water Pumps:**

They will be used if the moisture sensor indicates a low level of water in the soil. They will be pumping water from the main tank to the farming section.



Figure 55

- **Water Filter:**

It will be used for water filtration if the ammonia or any contaminants levels registered were high. This will be a safety precaution for both the fish and the plants, to avoid the pumped water being contaminated. This is a 220V module, so a relay will be attached to it to make it Arduino compatible.



Figure 56

- Lithium-Ion Batteries (Li-ion):

It will be used for power storage and providing a continuous DC current for any 12V component.



Figure 57

Merits:

Compatibility: Lightweight with a long lifespan. This makes it superior compared to mercury cells or dry cells.

Cost Effectiveness: It costs 70LE-100LE per cell, making it cheaper than $LiPO_3$ Cells or Lead-H_2SO_4 cells.

Demerits:

Aging Side-Effects: It is prone to losing some of its capacity as it ages.

Heat Sensitive: Under extreme temperatures (100 °C- 300 °C), it could spontaneously combust.

Mechanism Overview:

Using an Arduino Uno as a primary microcontroller to activate the actuator if the sensors indicate a change in the environment, and an ESP32 to send the data to a website that can be accessed remotely, making the project IoT-integrated.

The Chosen Parameters with their corresponding actuators:

- **pH**: Acid/Alkaline vials connected to a servo motor.
- **Water level**: An External water tank connected to a servo motor.
- **Soil Moisture**: A water pump connected to the main tank.
- **Turbidity**: A water filter attached to the Arduino via a relay.

Chapter III

(Constructing and Testing a Prototype)

I. Materials and Methods

II. Test Plan

III. Data Collection

I. Materials and Methods:

Materials:

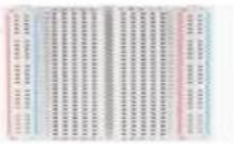
Plastic box	It will contain all of these		Turbidity sensor		To detect the clarity of the water of the fish tank
Jumber wires	To connect all parts		Water level sensor		To detect the level of water in the fish tank
Pumps	To take water from fish tank to plant		Ph sensor		To detect the PH level in the fish tank
Filter pumps	To filter the fish tank		Moisture level sensor		To detect the level of water in the plant
Relays	To control the power		Mint plant		The plant that we used
Stepdown volt	To take high volt and reduce it		Fish		The fish that we used
Breadboard	To connect all parts		Arduino		The mind of our prototype

Table 1

Methods:

The construction of the prototype (BioNexis) requires achieving design requirements and configuring the electric circuit. First, we make a schematic diagram of the project to simulate the prototype and its electrical connections

❖Steps of installation:

• Firstly, a deep search into the kinds of fish and plants. An 8-liter plastic box was selected for the fish and another box for plants.

• Research was done about the sensors to be used, like pH sensors, turbidity sensors, water level sensors, and moisture sensors. After these sensors were located, they were calibrated. Research was done to know how to connect the fish tank to plants with a pump to supply the plants with water and nutrients.

• A filter pump was used to clean the fish tank. Relay modules were used to control the electrical components, acting as switches. A step-down voltage regulator solved this problem. Jumper wires were used to connect these components to the breadboard.

• After a thorough search of the microcontrollers, Arduino was found to be the best option; it is the heart of the prototype. It is controlled by programming, in addition to the code being in AUTOSAR format, enabling the project to be compatible with bigger systems in the future. Finally, all components were connected to the plastic, and the prototype was ready for testing.



Figure 58

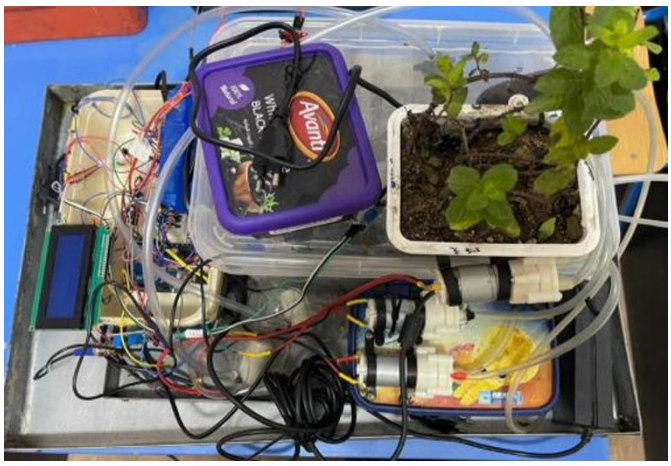


Figure 59

II. Test Plan:

A structured test plan was made to evaluate the results, which are (PH, moisture, water level, and turbidity). Each test was conducted for 6 continuous hours over the course of 5 days, using sensors, actuators, and a microcontroller.

• First: PH sensor output (Required Calibration):

As a part of our test plan, after the pH sensor was calibrated using buffer solutions of an know pH (3, 7, 11), after the rod was dunked in each of the buffers, the built-in potentiometer was twisted to match each of the corresponding pHs.

• Second, soil moisture sensor output (Pre-calibrated):

To start, the soil moisture sensor sends signals at a baud rate of 9600, which, in contrast to the rest of the components that operate at 115200, means that all upcoming results were displayed as sufficient or insufficient rather than in numerical values.

• Third, the water level sensor (Pre-calibrated):

As the water level sensor evaluates the water level using its exposed metal strips, combining this with the dimensions of the container (i.e, length and width), this will result in a continuous array that provides an accurate estimation of the amount of water in liters. If the water is less than 7L, a 12V water pump attached to a secondary tank resembling an external water source will refill the main tank.

• Fourth, the turbidity sensor (Pre-calibrated):

Turbidity is defined as Turbidity as a parameter is defined as water cloudiness by detecting how suspended particles scatter light at a 90-degree angle, which is referred to as a Nephelometric Turbidity Unit (NTU). As the turbidity sensor provides an analog output by evaluating how turbid the water is using an IR LED and detecting how much of the ray was diverted by the suspended particles. Turbidity. If it detects a turbidity of more than 40 NTU, the water filter will activate, filtering the water from the biohazards suspended in it.

III. Data Collection:

Tools:

Various tools were used to either measure or in the construction of the system, like:

Oscillating Saw:

The tool exhibited in Figure 60 is a saw that works using the concepts of oscillatory motion, as it only cuts hard objects, avoiding soft tissues and potential accidents.



Figure 60

Measuring tape:

It was used to measure the length of suitable wooden scraps that can be cut, as well as to measure the actual distance that the RC car covered during the course of the test plans.



Figure 61

Avometer:

It was used to determine the voltage and current intensity suitable for the components to avoid supplying them with more than they could endure.



Figure 63

Glue gun:

It was used to attach the wires, the sensors, and the jumper wires to the plastic container. This was done to avoid the components from falling off during the course of the system operation.



Figure 64

Screwdriver:

It was used in the extraction of the components, loosening them from their electronic boards. It was also used to fasten screws into the plastic container.



Calibrated pH Meter:

To further ensure the calibration process of the pH sensor, an already calibrated pH meter was fetched for the chemistry laboratory, with an error of ± 0.1.



Figure 67

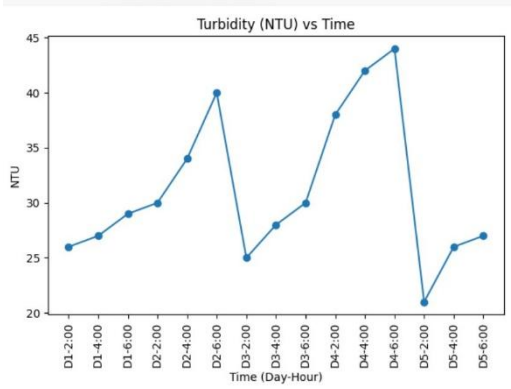
Results:

Day	Trials	Time	PH	Turbidity (NTU)	Water Level (L)	Moisture Status
Day 1	T1	10 am	7.5	26	8L	Irrigated
	T2	3 pm	7.5	27	8L	Needs replenishing
	T3	8 pm	7.4	29	7.8L	Irrigated
Day 2	T1	10 am	7.2	30	7.7L	Irrigated
	T2	3 pm	7.2	34	7.6L	Needs replenishing
	T3	8 pm	7	40	7.6L	Irrigated
Day 3	T1	10 am	6.8	25	7.5L	Needs replenishing
	T2	3 pm	6.5	28	7.4L	Irrigated
	T3	8 pm	6.5	30	7.4L	Needs replenishing
Day 4	T1	10 am	6.2	38	7.3L	Irrigated
	T2	3 pm	6.1	42	7.2 L	Needs replenishing
	T3	8 pm	6.3	44	8.1L	Needs replenishing
Day 5	T1	10 am	6.5	21	8.0L	Needs replenishing
	T2	3 pm	6.7	26	7.9L	Irrigated
	T3	8 pm	7.1	27	7.9L	Irrigated

Table 2

Negative Results:

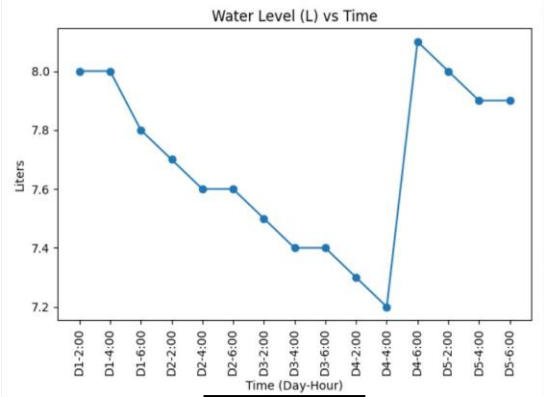
- As shown in Day 4 from 10 am to 3 pm, the turbidity rose to **44NTU from 38NTU**, at day 4 from 3 pm to 8 am, the water level sensor deviated from **7.2L to 8.1L**, and the pH values deviated from **6.7 to 7.1** as the actuator (acid pump) was activated to oppose the change.
- The PH sensor was not giving constant readings as shown in the table, so we made three trials for five days to get 6.8 ±0.5.



Graph 4

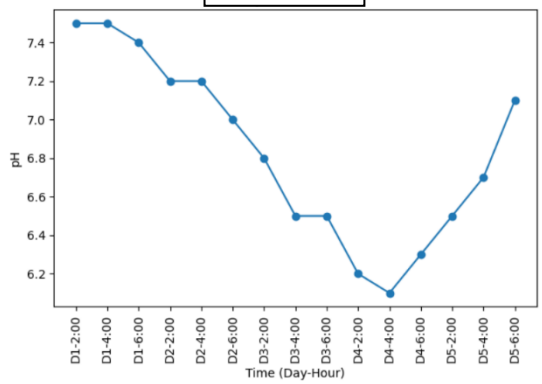
Positive Results:

- After solving the problems in the negative trial, which the PH sensor gives to the actuator, it regulates the water PH. The same applies to the water level sensor, and for the turbidity sensor, when it senses that water clarity reaches more than 40NTU and becomes constant for a time, the actuator doesn't receive the signal to clear the water. As the water clearness affected by living organisms, the turbidity sensor was trying to decrease the NTU as shown in the table to get an average value of 31.1 ± 0.5.



Graph 5

- As shown in the graph, the water was reduced and small rate, which gives us an average of 7.6 ± 0.4 L.



Graph 6

Chapter IV

(Evaluation, Reflection, Recommendations)

I. Analysis and Discussion

II. Recommendations

III. Learning Outcomes

IV. Conclusion

V. Citation

I. Analysis and Discussion:

In this section, the apparatuses used will be dissected, revealing their operating mechanism, and then the results will be analyzed to discover the correlation between them.

pH Sensor:

The pH sensor relies on the potential difference across the outer and inner surfaces of a glass electrode to measure the pH as follows:

The external reference electrode (first from left to right) is currently already integrated in the glass tube of the "combined" pH electrode.

The glass membrane electrodes for pH are mainly composed of a silicate matrix containing other atoms (e.g., Na, Li). In the glass membrane, the oxygen atoms of silicates are negatively charged, and this charge is compensated by Na^+ or Li^+ ions. The hydration increases the ion mobility, allowing an ion-exchange process with the H^+ ions of the sample, establishing a potential difference dependent on the H^+ activity in the external solution. As shown in Figure X.

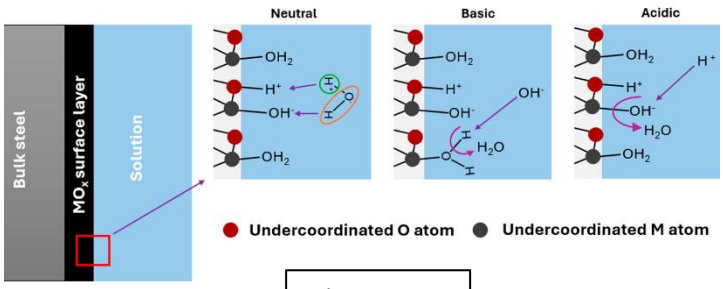


Figure 66

$$\Delta E = K + \frac{RT}{F} lna_H = K + 0.059\ loga_{H+}$$

In a potentiometric cell, several potentials are developed. There is potential associated with reference electrodes (E_{ref1} and E_{ref2}). For a pH electrode, the Nernst equation describes the potential as a function of the H^+ activity, according to Equation (1). ΔE is the potential difference (V), K is a constant (V), R is the ideal gas constant (8.314 J/mol K), T is the temperature (K), F is the Faraday constant (96,485.33 C/mol), and is the activity of the H^+ ions.

Turbidity sensor:

Turbidity as a parameter is defined as water cloudiness by detecting how suspended particles scatter light at a 90-degree angle, as shown in Figure 67.

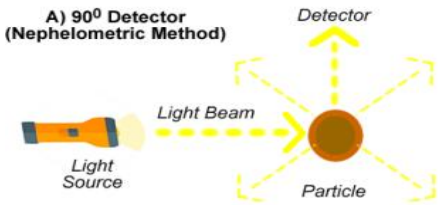
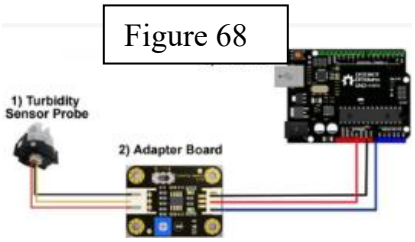


Figure 67

The sensor consists of: (1) a sensor probe; (2) the adapter board; and (3) an Arduino. The sensor probe houses the electronics for measuring the turbidity level (note that it is not waterproof in its original form). The adapter board contains an operational amplifier (opamp) comparator circuit adjust the trigger/threshold level for the digital output. The sensor operates at 5 V and returns an analogue value (between 0 and 1023) representing the sensor voltage on the Arduino's analogue pin (refer to the Arduino sketch). The sensor can operate in one of two modes: (1) analogue, or (2) digital outputs. In either mode, the sensor operates by sending a signal via an Infrared (IR) light-emitting diode (LED) across a solution to an IR phototransistor. All things being equal, the turbidity is approximated as a function of the reduction in voltage received by the IR phototransistor due to the scattering of IR light by particles (or light adsorption). The sensor measures in NTU (Nephelometric Turbidity Unit). Here are some examples of NTU values as presented in Table 3.

Figure 68



Filter grade	Turbidity (NTU)
Coarse	< 100
Coarse	< 80
Medium	< 60
Medium	12–35
Medium	10–20
Fine	1–10
Fine	1–2
Microorganism reducing	1–1.5
Microorganism reducing	1
Microorganism reducing	< 1
Microorganism reducing	< 1 ABSOLUTE

Table 3

Water Level sensor and Moisture sensor:

In contrast to the complexity of the pH and turbidity sensors' mechanisms, the water level and moisture sensors rely on the conductive properties of undistilled water, allowing an exposed metal strip to send signals to their boards, indicating the level of water.

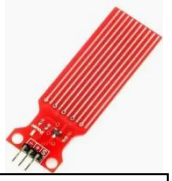


Figure 69

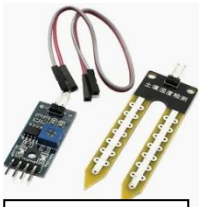


Figure 70

DC Water Pumps:

DC Water Pumps, whether 5V, 12V, or 24V, operate under the same fundamental principle, utilizing a DC motor to either propel or pull the water. DC motors operate on the physical principle of Lorentz force, where a current-carrying conductor placed in a magnetic field experiences a force: $\vec{F} = I\,\vec{dl} \rightarrow\times \vec{B}$ is the DC flowing through the conductor part with the infinitesimal length of $\vec{dl}$ is the magnetic flux density $\vec{B}$. This force results in the torque that causes the rotating part (rotor) to turn, resulting in mechanical motion.

The homogeneous magnetic field can be generated by windings supplied by DC or by magnets arranged around the rotating part. The part of the motor where these windings or the magnets are placed is called the stator.

A frame made of wire is placed in a homogeneous magnetic field, as shown in **Figure 71**. The current *I* is flowing through the wire of the frame. In this special case, when the vectors are perpendicular to each other, the Lorentz force equation has the simple form of *l* is the side length of the wireframe, and is the constant magnetic flux density amplitude. The Lorentz force is constant, and perpendicular to the magnetic field and the frame side, too.

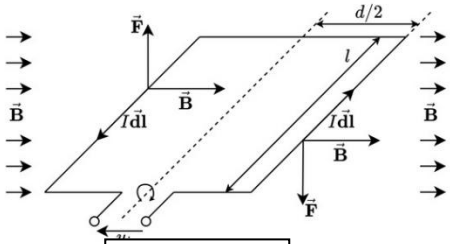


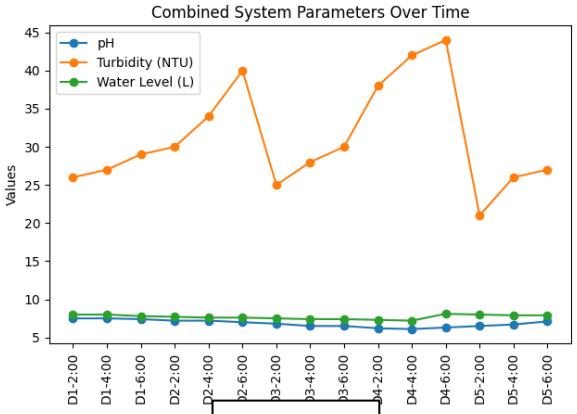
Figure 71

Result Analogy:

After conducting the test plans mentioned previously and gathering the data from 6 hours of continuous operation across 5 days, the following was found:

pH Analogy: The pH of the water changes steady rate, as there is a vacant volume above the water for the prawns to be able to emerge, as they are amphibious, meaning that CO2 is allowed to increase the acidity of the water, as $CO_2+H_2O \rightleftharpoons H_2CO_3$, then the carbonic acid partially dissociates to both $H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$ and $HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$ liberating at most 2 protons (H^+) from this reaction. This is opposed by the addition of NH_4OH to reduce the acidity. In contrast, the effect of the aquatic excretions resembled in ammonia is basic upon reacting with water, as it acts as a weak base in the reaction $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$, which liberates a Hydroxide group within the water. This is opposed by adding CH_3COOH to the main tank. The values registered were averaging at an acceptable rate of 0.3/h as shown in graph I. as there is a correlation between the water turning basic after the feeding times, tying them to the ammonia secretion. The rate of the pH shifting between acid and base was nearly 2/1 respectively, as the carbonic acid liberates at most 2 H^+ ions, while the ammonia liberates only one Hydroxide group.

Turbidity analogy: As the system is prone to experiencing turbidity due to the biological excretions from the aquatic prawns, a turbidity management system was installed for the purpose of keeping the Nephelometric Turbidity Unit (NTU) values below the threshold of 40. On average, the turbidity reached the threshold on day 4 of the operation, noting that the water was replaced via the replacement pump before conducting the test. The rate of turbidity increase can be visualized as shown in graph II. On day 4, for example, the turbidity exceeded the limit at 6:00 pm in T3 of the operation. Graph II: Turbidity values vs Time. After reaching the limit, the filter reduced it to a reasonable 20NTU after 2 hours. In case the filter malfunctions or the water is too turbid (>40NTU), there is an emergency mechanism that replaces the



Graph 7

water. The replacement pump empties the tank of the turbid water, and the refill pump fills it back in about 2 minutes.

Moisture Analogy: As mentioned in the methods, the moisture sensor operates at a baud rate of 9600, whilst the rest of the components operate at 115200, which makes the whole system prone to malfunctions during initialization if they interact with the sensor, like the LCD displaying gibberish whenever it tries fetching readings from it. So, instead of displaying numerical values, the status will be displayed. Phrases like “needs replenishing”, “irrigated”, and “Overhydrated” indicate how hydrated the plant is. The threshold is established in the Arduino, as it operates at different baud rates and ADCs simultaneously, and the actuator and the status displayed are sent from it. The plant needs only 2 minutes to be irrigated.

Water Level Analogy: As mentioned in the pH section, the water level is at 12 cm from the bottom, making the volume of water in the tank about 8L, as the length(28.5cm) and width(19cm) are multiplied. This level could fluctuate according to multiple factors (e.g., temperature increase and humidity). This, however, can be controlled by the 12V refill pump to maintain the level in the event of fluctuations. The effect is insignificant until the mode is changed for the database organism (freshwater shrimp), as they require the whole tank to be filled (10L of water). The sensor has a conductive length of 5cm, meaning the 5cm needs to be submerged underwater to signal the pump to stop.

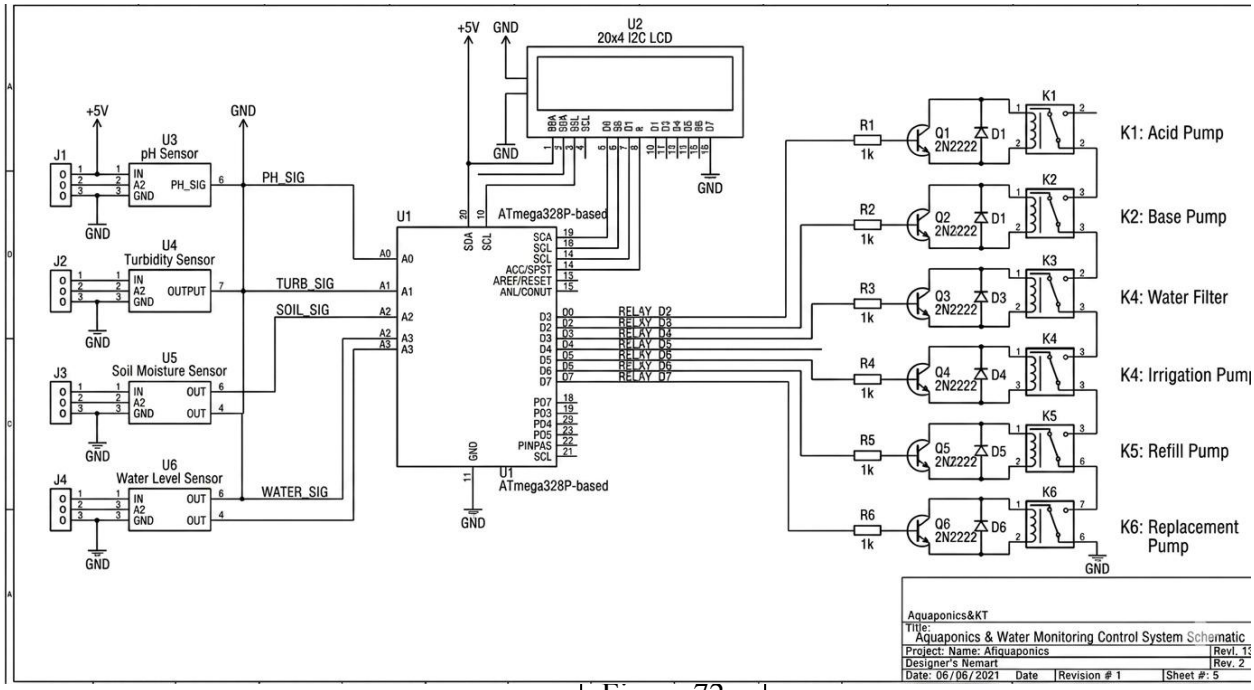


Figure 72

Power consumption: The project operates using a 12V adapter attached to 2 buck converters, 9V and 5V, as shown in Figure 72. After calculating the worst-case power consumption, the result was 30 W, which is much lower than your blender operating at max settings, which requires 1000 W. Moreover, using this formula, the energy efficiency is ~93%, as the total input is 30W and the useful output is 28W.

Estimated Annual Yield: Using the compound increment equation, a solid estimation of the increase in population N0 is the initial number, r is the rate of reproduction, d is the mortality rate, m is the number of mating cycles, and t is the number of seasons. Plugging the corresponding data into the equation, it is discovered that starting with only a pair results in a 700% increase in population. ‘Equation III: Compound Incrementation

II. Recommendations:

If the project is to be replicated at an agency-level plant, some adjustments and further upgrades are a necessity to achieve the optimal outcome possible. These upgrades are:

- **Improving circulation:** Instead of using plastic tubes, metal ones will be more durable and will negate malfunctions with less maintenance cost. However, if the recycling trait is put in perspective, using durable PLA or PVC pipes could theoretically have the same effect, but with more maintenance costs, unfortunately.

- **Reducing External Power Reliance:** Instead of using all the power from external power infrastructure or commodities to lift water, for example, generating a part of it via attaching water turbines to the pipelines could reduce the operation cost, making the profitability much higher and a viable option for future investors.

- **The Implementation of Nitrobacter:** Aquatic creatures not only secrete ammonia and nitrates in their excretions, but also nitrites, which are poisonous for plants. Utilizing Nitrobacter to break down the nitrites in the pumping tubes or installing a nitrite filtering station will ensure the plant's well-being.

- **The Addition of More Parameters:** The addition of more sensors or parameters will ensure the organisms’ well-being, as parameters like temperature, salinity, or dissolved oxygen will facilitate estimating the status of the controlled environment.

III. Learning outcomes:

Learning Transfer

BI.09	**Introduced the concept of negative/positive feedback, as the human body is full of them, such as the ovarian/menstrual cycles, and the spermatogenesis.**
MA.06	**Allowed the deduction relations between the parameter changing rates with respect to time, as the derivatives were introduced in this Learning outcome.**
MA.09	**Allowed the integration of the motor power consumption function of time to calculate the pump's overall power consumption.**
MA.10	**Facilitated the calculation and laid an accurate estimation for the annual yield of the project, using the formula of compound incrementation.**
ME.05	**Introduced the concept of power, enabling the calculation of the power consumed by the prototype and upscaling it to suit the larger module/applications.**
PH.11	**Opened the possibilities of reducing/increasing the voltage and amperage, as the concepts of transformers, buck converters (step-up/down) were taught.**
PH.12	**Explained the mechanism behind the DC motors used as pumps and how to maximize their efficiency.**
PH.13	**Introduced the concept of LCR circuits and how they can be utilized in our project.**
PH.15	**Facilitated the usage of a rectifier to convert the AC to usable DC for the prototype.**

Table 4

IV. Conclusion:

Five days of test results collected during the outside test confirmed that stable environmental parameters were kept in the range appropriate for prawn life and plant growth. The data obtained shows that the pH value fluctuated between 6.1-7.5 in 15 measurements, turbidity varied from 25 to 42 NTU, while the water level was kept in the range of 7.1-8.1 L. Also, irrigation of the substrate was automatically carried out whenever it was necessary due to lack of moisture. The closed loop control system worked well by providing feedback control based on sensor and actuator calibration.

All Capstone project design requirements were met. There were no pauses exceeding six hours of work in a row, recycled plastic bottles were used in the prototype, and there was always a possibility of automatic regulation of environmental parameters. The nutrient recycle function was activated; thus, saving water and chemicals, which contributed to resolving some of Egypt's Grand Challenges, such as garbage recycling and waste reduction, expanding the area of agricultural lands, pollution reduction, and managing population growth and the demand for food. Thus, test results confirm that the developed inexpensive, but highly automated prototype of the aquaponic system is effective, practical, and ready to go on display. This model can demonstrate how simple feedback controls made of waste products can help create an aquaponics system in any city like Alexandria. The prototype meets all requirements to be scaled for future farming in Egypt.

Moreover, after ensuring the execution of the design requirements, generating a more preferable solution, ensuring the well-being of both the plants and the aquatic creatures, it can be concluded that this project was a success by all means.

V. Citation:

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